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Master Thesis

Beyond EER: Speaker-Level Evaluation and Performance Assessment of Voice Anonymization with Alternative Metrics and Phonetic Features Analysis

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Ioanna Karagianni, Stuttgart, July 2026

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Abstract

Voice anonymization systems are conventionally evaluated using the Equal Error Rate (EER), a single population-level threshold that has been shown to obscure substantial variability in privacy protection across individual speakers. This thesis reproduces three baseline systems from the 2024 Voice Privacy Challenge (VPC) i.e. STTTS, NAC, and ASRBN and evaluates them not only with global and speaker-level EER, but also with two alternative frameworks: The evidence-based ZEBRA framework (dECE and worst-case LLR computed at the speaker-level) and the legally motivated Linkability & Singling Out metrics. In addition, a feature-level analysis examines whether phonetic, prosodic, acoustic, and linguistic properties preserved after anonymization are associated with individual speaker vulnerability. Results show that while ASRBN achieves the strongest population-level privacy protection under both EER and ZEBRA, a subset of speakers remains consistently re-identifiable across all three architectures, confirming that global metrics can mask persistent individual-level failures. Linkability & Singling Out reveal that system rankings shift once evidence is aggregated across utterances: NAC, ranked second under EER and ZEBRA, shows the highest residual Linkability at longer conversation lengths, showing that low verification error does not guarantee resistance to re-identification. The feature analysis provides exploratory evidence that rhythm, phone-duration, and pitch/formant trajectories retain speaker-discriminative structure and show negative descriptive associations with speaker-level EER in several conditions, while linguistic similarity adds complementary predictive power. These findings support combining population-level, speaker-level, and real-life grounded metrics over global thresholds alone, and point to residual rhythmic, durational, acoustic, and linguistic cues as targets for more robust anonymization.

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1 Introduction

In the last years, speech data seem to be in high demand, due to the magnitude of applications they can be utilized for. From voice assistants and smart home systems, psychotherapy and medical consultations, to voice communication in IoT (?) the need for big amounts of speech data to train Speech Synthesis and Speech Recognition Systems is high. This need also brings some privacy concerns in the way these data are collected, stored and utilized as speech recordings contain personally identifiable information and not only explicit identity markers such as the speaker’s low level descriptors (pitch, loudness, jitter etc.), but also demographic, biometric, and emotional attributes that can be inferred from the signal itself (?).

Consequently, the use of raw speech data raises concerns regarding re-identification and unintended inference, especially under data protection frameworks such as the **EU General Data Protection Regulation (GDPR)**, which defines anonymous information as:

“... anonymous information, namely information which does not relate to an identified or identifiable natural person or to personal data rendered anonymous in such a manner that the data subject is not or no longer identifiable.” (?)

This anonymization does not only require the alteration of the speaker’s voice, in order to make them non-identifiable, but also the removal or the change of specific words in an utterance and sounds in the background (?). Therefore, to address these risks and to develop systems that comply with the privacy laws, the **Voice Privacy Challenge (VPC)** series (2020–2024) has emerged (???). **Voice Privacy** comprises the set of transformations applied to a speech signal to remove, mask, or obfuscate speaker-specific and sensitive cues (such as gender, accent, emotion, or health signals) that enable re-identification or inference of the speaker, while maintaining utility of the speech (e.g. paralinguistic, linguistic content) for downstream tasks (?). The goal of the challenge is to support the participants to advance and

evaluate techniques for voice anonymization (??). The term **Voice Anonymization** refers to the transformation of a speech signal such that the speaker’s identity is concealed while at the same time the linguistic content and the paralinguistic attributes of the utterance are preserved (the so-called: **Privacy-Utility tradeoff**) (?).

During the last years, there have been remarkable developments in the anonymization systems developed by the VPC organisers and the participants (???) and a transition from traditional models in the first two challenges (??) to neural, more advanced ones in the latest VPC of 2024 (??). However, the development of these systems underlines that anonymization is not only a technical challenge but also a question of how privacy should be quantified, measured and safeguarded in real-life scenarios. Ever since the beginning of the Challenge in 2020, doubts are openly expressed regarding the credibility of the standard privacy metric utilized consistently across all the VPCs, which is the **Equal Error Rate (EER)**. These doubts were centered around the inability of this metric to capture and express the variability of privacy protection across the individual speakers, as it is noticed that the anonymization performance of some individual speakers remains very low across fundamentally different anonymization systems (???). These limitations of the EER have motivated speech engineers to explore alternative or complementary evaluation metrics, like **Linkability & Singling Out** (?) grounded in data protection laws and the **Zero Evidence Biometric Recognition (ZEBRA) framework** inspired by forensics (??).

Regardless of the need for different evaluation metrics, the phenomenon of some speakers performing low in the EER evaluation, even though different anonymization systems have been utilized, persists and requires further attention and investigation (??). More specifically, there seems to exist the hypothesis that there are speakers that are easier to anonymize (resulting in higher EER scores) and speakers that are harder to anonymize (resulting in very low EER scores). This trend can be attributed to a number of factors, like the linguistic content of the speaker’s speech and the temporal dynamics of their speech (??). Despite these first explanation efforts on the analysis of the individual speakers’ performance, a collective targeted work providing

an in-depth study of the prosodic, acoustic and phonetic characteristics is yet to be performed.

In this thesis, the primary focus falls on the examination of alternative evaluation metrics for assessing privacy in voice anonymization systems, as proposed in recent literature and on providing a deeper understanding of how these metrics capture privacy preservation compared to the traditional EER used in the VPC 2024. Moreover, an effort is made to investigate the phonetic, acoustic and prosodic factors and their combination with linguistic factors at the speaker level, that may affect negatively the effectiveness of anonymization. To address these initial hypotheses, a series of experiments is conducted using the baseline systems, the attacker model and evaluation pipeline provided by VPC 2024 (?), in combination with alternative evaluation frameworks and metrics introduced by the earlier and latest research in the field.

Therefore, a gap can be identified from which four main research questions (RQs) seem to emerge, offering a ground for further experimentation and research:

RQ1: How well or poorly do the three best-performing anonymization systems in the Voice Privacy Challenge of 2024 (B3, B4, B5) preserve privacy across individual speakers, and where are their differences noticed?

RQ2: How do other existing metrics (Linkability & Singling Out, ZEBRA) seem to perform after evaluating the same baseline systems, for the same speakers?

RQ3: Do some other factors, such as gender, phoneme duration per speaker, pitch, rhythm, formant trajectories, and the linguistic content of speech, seem to affect the anonymization systems' performance?

RQ4: What could be some weaknesses of the latest Voice Privacy Challenge anonymization systems, based on the findings of the evalu-

ation metrics used and the feature analysis conducted, and what should be taken into consideration to make them more robust?

To answer these questions, this thesis reproduces the STTTS, NAC, and AS-RBN baselines of VPC 2024 and evaluates them using global and speaker-level EER (RQ1), the ZEBRA framework, and the Linkability & Singling Out metrics (RQ2). In addition, phonetic, prosodic, acoustic, embedding-based, and linguistic features as well as the role of gender (RQ3) are analyzed to examine which preservation properties remain speaker-discriminative after anonymization and whether they are associated with speaker vulnerability. Lastly, the findings of the evaluation metrics and the feature analysis are discussed to identify weaknesses of the anonymization systems and provide recommendations for future implementations (RQ4).

The results demonstrate the importance of evaluating voice anonymization from multiple perspectives. Although the systems provide meaningful privacy protection at the population level, some speakers remain consistently vulnerable across different anonymization architectures. Moreover, the system rankings depend on the evaluated threat, particularly when several utterances from the same speaker are aggregated. The feature analysis further indicates that rhythm, phone-duration patterns, pitch and formant trajectories, as well as linguistic similarity with the enrollment utterances may preserve residual speaker information, although their contribution varies across systems and trial partitions. These findings support the use of alternative population-level and speaker-level frameworks that underline different aspects of leakage rather than relying on a global threshold-based metric alone.

The remainder of this thesis is organized as follows: Section ?? introduces the VPC and its components. Section ?? reviews related work on the limitations of EER, alternative privacy metrics, and residual phonetic and linguistic identity cues. Section ?? describes the experimental methodology, followed by the results in Section ?? and their discussion in Section ?. Section ?? concludes the thesis and outlines directions for future research. The code used to reproduce the results reported in this thesis is available in the following GitHub repository:

<https://github.com/joannakarayianni/Voice-Privacy-Challenge-2024-Thesis>

2 Background Work

2.1 The Voice Privacy Challenge Initiative

The integration of speech systems into numerous services and industries has made speech data a valuable resource for training reliable speech systems (?). Yet speech data is also sensitive, as it inherently encodes cues about a speaker's identity (e.g. gender, age, health indicators). The GDPR, which emerged in 2016 (?), recognises biometric data used to uniquely identify a natural person as a special category of sensitive personal data (Article 9.1), whose processing is in principle prohibited unless specific legal exceptions apply (Article 9.2). For speech data the risk of privacy intrusion arises from the moment speech is captured and persists throughout its processing and storage lifecycle. Moreover, the multifunctionality of speech data increases the risk of "function creep," where data collected for one purpose (e.g. speech recognition) may later be used for identity recognition or profiling without legal basis (?).

Speaker anonymization emerged from these challenges as a key technical strategy to reduce speaker identifiability while keeping speech data usable for downstream tasks (Article 5.1 - "data minimization"), motivating the VPC initiative in 2020. Its goal is to bring together researchers, speech engineers, and privacy professionals to develop state-of-the-art anonymization systems (?).

The organisers describe privacy preservation as a "game" between users who share data for legitimate downstream tasks and an attacker who tries to infer speaker identity from that data. An anonymization system is applied to the speech signal, while an Automatic Speaker Verification (ASV) system acts as the attacker, assumed to have knowledge of the anonymization mechanism depending on the attacker setup defined per challenge version. The organising committee provides common baseline systems, fixed training/development/evaluation datasets, and standard evaluation protocols for fair comparison. Systems are evaluated jointly on privacy and utility, since the linguistic and paralinguistic value of the data must be preserved. VPC has therefore established a shared experimental framework in an effort to align technical

progress with real-world privacy requirements.

2.2 Baseline Anonymization Systems

Teams are provided with baseline anonymization systems of various architectures to enhance for optimizing the privacy-utility tradeoff under the common evaluation protocol. The following sections review these baselines, focusing on VPC 2024.

2.2.1 Voice Privacy Challenge 2020

The first edition introduced two baselines. **B1** is based on x-vector anonymization with neural speech synthesis, following a three stage pipeline of feature extraction, x-vector anonymization, and speech synthesis. Speaker identities are represented as x-vectors extracted via a TDNN model, linguistic content as bottleneck (BN) features from a TDNN-F ASR model, and F0 via a standard pitch tracker. The new anonymized x-vector gets computed by averaging candidate x-vectors from a speaker pool using Probabilistic Discriminant Analysis (PLDA), defining a pseudospeaker, after which an acoustic model synthesized Mel-spectrograms from the anonymized x-vector, original BN features and original F0 (?). **B2** uses traditional signal processing via the McAdams coefficient, decomposing speech with Linear Predictive Coding (LPC) and modifying the angles of complex-conjugate poles before resynthesis (?). Of the 25 submitted systems, 16 were successful, with B1 offering better anonymization results, though at the cost of utility degradation (??).

2.2.2 Voice Privacy Challenge 2022

VPC 2022 retains and enhances the 2020 paradigm. **B1.a** kept the original synthesis backend with a Neural Source Filter (NSF) vocoder, while **B1.b** converted BN, F0, and x-vector features directly using an NSF model with a HiFi-GAN vocoder (?). **B2** remained unchanged. Among 16 valid submissions from 6 teams, most outperform the baselines under the semi-informed attacker, but no system achieved consistently

optimal performance as stronger privacy protection consistently came at the cost of utility and voice distinctiveness (??).

2.2.3 Voice Privacy Challenge 2024

VPC 2024 introduced six baselines, with **B1** and **B2** being retained from 2022. The three new baselines evaluated in this thesis represent fundamentally different anonymization paradigms. **B3** (?) marks a shift toward ASR+TTS style anonymization: Speech is factorized into discrete phonetic transcriptions and prosodic attributes, a Wasserstein GAN generates pseudospeaker embeddings explicitly dissimilar from the original via a cosine threshold, and pitch and energy are randomly perturbed at the phone level to suppress fine-grained speaker-specific prosodic patterns while preserving the global prosodic structure (?). Then, **B4** treats anonymization as a conditional sequence generation problem in token space: Semantic tokens from the source utterance are concatenated with acoustic tokens from a pseudospeaker pool, and a GPT-like decoder generates new acoustic tokens satisfying both semantic consistency and pseudospeaker style constraints (?). Lastly, **B5** and **B6** apply vector quantization (VQ) to Automatic Speaker Recognition (ASR) bottleneck features to reduce speaker leakage at the feature level, with B5 using a pretrained wav2vec2 model followed by Time Delay Neural Network (TDNN-F) layers and B6 using a fully supervised TDNN-F architecture, allowing assessment of whether self-supervised pretraining introduces stronger speaker leakage (?).

Of the 40 registered teams, 13 submitted a total of 36 systems, yielding influential contributions (?). ? adopted a disentangled neural codec with step-by-step disentanglement of speaker identity, linguistic content, and emotional state, achieving an EER of 36.99%, WER (Word Error Rate) of 1.85%, and UAR (Unweighted Average Recall) of 64.48% on the test set. ? proposed a system combining WavLM feature extraction with feature interpolation and explicit sentiment transfer, achieving EER scores of 34.12% (female) and 36.08% (male), a WER of 2.37%, and a UAR of 60.95%, obtaining the best privacy-utility-emotion tradeoff across the top three privacy categories (??).

2.3 Attacker Systems

The second fundamental component of the VPC is the attacker system, in the form of an ASV model, which constitutes the threat scenario under which anonymization systems are evaluated. Its goal is to re-identify the original enrollment speaker from the anonymized trials.

VPC 2020 defines three attacker models depending on the adversary’s level of knowledge (??): the **ignorant attacker**, unaware that anonymization was applied, who verifies anonymized trial utterances against original enrollment utterances using an ASV system trained on original data. Then the **lazy-informed attacker**, who knows the anonymization method but not the pseudospeaker mapping, and operates on anonymized enrollment and trial utterances with an ASV system trained on original data, and lastly the **semi-informed attacker**.

The semi-informed attacker constitutes the strongest threat model and was adopted as the default setting in VPC 2022 and VPC 2024 (??). In both editions, the attacker has full knowledge of the anonymization system and access to multiple enrollment utterances per speaker, anonymizes enrollment data using the same pipeline as the trial data, and re-trains the ASV system accordingly. The only difference in VPC 2024 is that anonymization operates at utterance level, meaning pseudospeaker assignment is performed independently per utterance rather than per speaker (?).

2.4 Evaluation Metrics

In this section the evaluation metrics used in the assessment of voice anonymization systems are presented, organized by challenge edition.

2.4.1 Voice Privacy Challenge 2020

VPC 2020 (???) follows an extensive evaluation framework combining **objective** and **subjective** metrics, with objective metrics derived from ASV and ASR systems and subjective metrics based on human listening tests.

Equal Error Rate (EER). Originally a biometric authentication metric (?), the EER measures verifiability after anonymization. A threshold θ determines whether two utterances are classified as belonging to the same speaker, with two possible error types: a *false alarm* (two different speakers incorrectly accepted) and a *miss* (two utterances from the same speaker incorrectly rejected):

$$(1) \quad \begin{aligned} P_{\text{fa}}(\theta) &= \frac{\#\{\text{impostor trials with score} > \theta\}}{\#\{\text{total impostor trials}\}}, \\ P_{\text{miss}}(\theta) &= \frac{\#\{\text{target trials with score} \leq \theta\}}{\#\{\text{total target trials}\}}. \end{aligned}$$

The EER corresponds to the point where both error rates are equal:

$$(2) \quad \text{EER} = P_{\text{fa}}(\theta_{\text{EER}}) = P_{\text{miss}}(\theta_{\text{EER}}).$$

A low EER indicates accurate speaker recognition and therefore weak anonymization, while a high EER (above 30%) reflects stronger anonymization and higher privacy.

Log-likelihood-ratio cost functions. As noted by ?, C_{llr} is an application-independent metric measuring the cost of soft detection decisions over all operating points:

$$(3) \quad C_{\text{llr}} = \frac{1}{2} \left[\frac{1}{N_{\text{tar}}} \sum_{i \in \text{targets}} \log_2(1 + e^{-\text{LLR}_i}) + \frac{1}{N_{\text{imp}}} \sum_{j \in \text{impostors}} \log_2(1 + e^{\text{LLR}_j}) \right],$$

where N_{tar} and N_{imp} are the number of target and impostor trials respectively. A perfectly discriminative system achieves $C_{\text{llr}} = 0$, while $C_{\text{llr}} = 1$ corresponds to random attacker performance and perfect anonymization (?). C_{llr} decomposes into a **discrimination loss** $C_{\text{llr}}^{\text{min}}$, reflecting target-impostor separation under optimal calibration, and a **calibration loss**, measuring how well raw scores are already calibrated (?). Higher values indicate stronger privacy protection.

Voice Similarity Matrices: DeID & GVD. Introduced by ? and used as post-evaluation in VPC 2020 (???), these metrics visualize anonymization performance

across speakers via a voice similarity matrix $M = (M(i, j))_{1 \leq i, j \leq N}$ built from LLR scores:

$$(4) \quad M(i, j) = \sigma \left(\frac{1}{n_i n_j} \sum_{k=1}^{n_i} \sum_{l=1}^{n_j} \text{LLR}(x_k^{(i)}, x_l^{(j)}) \right),$$

with three variants computed: M_{oo} (original-original), M_{aa} (anonymized-anonymized), and M_{oa} (original-anonymized). Speaker separability is measured via diagonal dominance:

$$(5) \quad D_{\text{diag}}(M) = \left| \frac{1}{N} \sum_{i=1}^N M(i, i) - \frac{1}{N(N-1)} \sum_{\substack{j,k=1 \\ j \neq k}}^N M(j, k) \right|.$$

De-identification (DeID) measures how well anonymized voices are dissociated from their originals:

$$(6) \quad \text{DeID} = 1 - \frac{D_{\text{diag}}(M_{oa})}{D_{\text{diag}}(M_{oo})},$$

where DeID= 100% indicates perfect de-identification. **Gain of Voice Distinctiveness** (GVD) evaluates whether anonymized voices remain mutually distinguishable:

$$(7) \quad \text{GVD} = 10 \log_{10} \left(\frac{D_{\text{diag}}(M_{aa})}{D_{\text{diag}}(M_{oo})} \right),$$

where GVD= 0 indicates preserved distinctiveness, with positive and negative values corresponding to a gain or loss respectively.

Word Error Rate (WER). WER assessed utility via ASR performance on anonymized speech and was retained as the primary utility metric across all three challenge editions.

Subjective evaluation assessed both privacy and utility (??). For privacy, **Speaker Verifiability** rated original-anonymized utterance similarity on a 1-10 scale, while **Speaker Linkability** assessed how distinctively human listeners perceived different speakers' voices. For utility, **Speech Intelligibility** and **Speech Naturalness** were each rated on a 1-10 scale via crowdsourced evaluation.

2.4.2 Voice Privacy Challenge 2022

VPC 2022 retained the 2020 evaluation framework with refinements (??). The key novelty was a **Ranking Policy**: rather than ranking systems globally, four minimum EER thresholds (15%, 20%, 25%, 30%) defined separate evaluation conditions, with systems meeting each threshold ranked by average WER. This allowed comparison within comparable privacy bands and confirmed that higher privacy consistently came at the cost of utility (?). EER, WER, and GVD were used unchanged from VPC 2020 (see Section ??), with one new metric introduced.

Pitch Correlation Metric - ρ^{F_0} . This new utility metric assessed whether anonymization preserved the intonation contour of the original utterance. Pitch contours are extracted for both utterances, the shorter sequence is linearly interpolated to match lengths, temporal lag alignment is applied, and the Pearson correlation over voiced segments is computed. All submissions were required to achieve $\rho^{F_0} > 0.3$.

2.4.3 Voice Privacy Challenge 2024

VPC 2024 retained only objective metrics (?). Here, the ranking policy from VPC 2022 was retained with adjusted EER thresholds (10%, 20%, 30%, 40%), with systems meeting each threshold ranked jointly by WER and UAR.

EER remains the sole **privacy** metric (see Section ??), computed using an **Emphasized Channel Attention Propagation and Aggregation Time Delay Neural Network (ECAPA-TDNN)** ASV system under the semi-informed attacker model (?).

For **utility**, **WER** is computed using a fixed ASR model trained on original LibriSpeech-train-960 (see Section ??). The newly introduced **UAR (Unweighted Average Recall)** expands utility evaluation to emotional content preservation, computed by a SER (Speaker Emotion Recognition) system:

$$(8) \quad \text{UAR} = \frac{\sum_{i=1}^{N_{\text{class}}} R_i}{N_{\text{class}}},$$

where N_{class} is the number of emotion classes and R_i is the per-class recall. Higher values indicate better emotional content preservation (?).

3 Related Work

3.1 The Problem with EER

Recent studies have shown that the EER does not accurately capture the variability of privacy protection across speakers. It has been reported that anonymization performance for an individual can still remain very low or very high even though fundamentally different anonymization architectures are employed. In other words, the global EER values often mask substantial inconsistencies in the speaker level and may give a misleading impression of the actual anonymization system’s performance.

? argue that anonymization systems exhibit strong speaker-dependent behaviour and identify speakers who are consistently either easy to imitate or difficult to recognize, regardless of the anonymization method used. Their analysis reveals that many speakers maintain similar privacy characteristics across all systems, and accordingly they are being categorized in a rank from easy to difficult to recognise. This means that some speakers remain vulnerable across different anonymization methods, showcasing weaknesses in evaluation datasets and in the anonymization process itself.

In ?, their analysis begins by running the B3, B4, and B5 anonymization systems from the VPC 2024 and computing individual per speaker EER scores under the semi-informed attack model. This reveals that while mean EER values fall between 20-35 %, several speakers across both genders and across the three baselines score extremely low EERs (even 2%), indicating that anonymization repeatedly fails for them across all systems. Motivated by these findings, the authors wanted to check if the vulnerabilities of some speakers arise from the linguistic patterns they follow. They test this by adapting a BERT model as a text-based ASV attacker applied directly to the transcripts of anonymized utterances, with this resulting in EERs as

low as 1–2% for the same vulnerable speakers. When trying to investigate the linguistic patterns used by these individuals they found that characteristic vocabulary and thematic patterns, such as religious terms, were able to create intra-speaker linguistic similarity that attackers can exploit. That means that, some speakers can be re-identified purely from textual information even when their acoustic profiles have been anonymized. Following this study, ? demonstrate that the used evaluation corpus in VPC 2024, LibriSpeech contains significant content leakage, as the speakers’ utterances are constituted by them reading books with distinct lexical and phonetic distributions. As a result, speaker verification systems are shown to once again partially rely on differences in phone inventories (meaning specific word usage) rather than purely on biometric voice characteristics. From their experiments using representations based solely on phone identities and phone durations, they show that a non-trivial re-identification performance can still be achieved even when spectral information is removed or altered through a TTS-ASR anonymization system. Also, they report a strong correlation between speaker-specific phonetic distributions and EER (a Pearson correlation of 0.59), indicating that content variability itself acts as an identity cue.

Also, EER’s unreliability may not stem only from speaker-dependent variability, but also from confounds built into the evaluation protocol itself. ? show that the standard VPC privacy evaluation overestimates privacy when a same-gender target selection algorithm assigns pseudospeakers, because the attacker’s recognizer learns to encode target-speaker information rather than source-speaker identity alone. When correcting this with an adversarial, gradient-reversal classifier they noted lower, more realistic EER estimates. This suggests that how pseudospeakers are assigned during attacker training can itself distort EER-based privacy claims.

In another related study, ? argue that apart from the fact that EER is average-based (which is already mentioned), it can obscure adversaries who succeed with high confidence against only a subset of speakers. Reframing speaker privacy as a membership inference problem, they show that anonymization systems with near-identical EER can differ by orders of magnitude in true positive rate once the adversary is evaluated at very low false-positive rates. This means that a system can appear

equally private under EER while in fact permitting high-confidence re-identification of specific individuals. When applying this analysis directly to the VPC 2024 AS-RBN baseline, they also find that simple learned attackers recover speaker gender and accent with near to perfect accuracy even after anonymization, suggesting that EER evaluation can mask the leakage of sensitive demographic attributes that are not captured by EER.

All in all, these findings reveal a fundamental limitation of global EER: As a single aggregating score it cannot expose the individuals who become outliers, performing either very poorly, nor the sensitive attributes that may remain identifiable even when speaker identity itself is well protected.

3.2 Alternative Metrics Focusing on Individual Speakers

These limitations of the EER have motivated speech researchers to explore alternative evaluation metrics, like **Linkability & Singling Out** (?) which are grounded in data protection laws, and **Zero Evidence Biometric Recognition (ZEBRA)** which is inspired by forensics and aims to understand the variation in privacy preservation provided to a population (?).

The ZEBRA framework measures the gap in the attacker system’s prior belief and posterior belief after it has observed the anonymized utterances. It provides two complementary metrics, the first being the **Expected Privacy Disclosure** (calculated through Empirical Cross Entropy - ECE) which reflects the amount of identity evidence remaining across the whole population (??). As it is calculated in bits, 0 equals a perfect anonymization. To be more precise, it is computed as the area between the ideal ECE curve and the observed curve integrated across all the priors. A perfect anonymization equals a likelihood rate (LR) of 1 (meaning no information gain for the attacker system).

The other metric in the framework is the **Worst-case Privacy Disclosure** which captures the strongest piece of evidence for an individual speaker that was able to survive after the anonymization has been performed. It is the maximum absolute log-likelihood ratio (LLR) observed when testing the system, which is then

mapped into an A-F categorical ranking ("A" showing best privacy). If LLR equals 0 then that means perfect anonymization. All in all, the ZEBRA metric seems ideal for evaluation, as it is not based on a single global threshold that ignores outliers, but also provides information about the individual levels of achieved privacy.

In a similar manner, **Linkability** measures the probability that an anonymized trial embedding is more similar to its true enrollment embedding than to any other speaker's embedding from the pool of enrollment speaker embeddings. High Linkability means therefore that although the speaker identity might not be directly revealed, an attacker can aggregate multiple anonymized samples and reconstruct a stable voice profile for an individual speaker. This is again considered a failed anonymization because the privacy law says that anonymized data should not allow records of same speakers to be connected across different contexts.

Singling Out, estimates the probability that an attacker can design a binary predicate that "fires" on exactly one speaker's anonymized representation. Even if the attacker cannot identify who the speaker is, just isolating a single entry can already be thought of as a failed attempt to keep privacy. It is also noteworthy that Linkability & Singling Out are suggested by the organisers of the VPC of 2024 as privacy metrics that should be integrated in the evaluation pipeline for future versions of the challenge (?).

3.3 Prosodic Features as Indicators of Speaker Identification

3.3.1 Prosody and Pitch Trajectories

It is well known that prosody in speech plays a major role in distinguishing speakers from one another in perceptual experiments. Prosody is not a single acoustic characteristic, rather a combination of F0 (fundamental frequency or pitch) contours, the durations of segments and pauses in speech (i.e. speech tempo) and the specific intonation or linguistic syllable stress. All of them are features that convey an underlying emotion or specific message (??). An individual's pitch characteristics have been proven to contain cues about their identity, as they are based on physiological

properties such as the shape of the vocal tract and the larynx. Therefore, speech scientists in the field of voice anonymization have developed systems that target the concealment and modification of the original speaker F0 trajectories. In many voice anonymization systems, speaker identity is represented through x-vector embeddings extracted from the source speech, while pitch trajectories, were often preserved or modified separately during synthesis (?). In this context, ? tested a speaker de-identification approach that explicitly targets dynamic F0 trajectories using Functional Data Analysis (FDA) and showcased that pitch variation can be decomposed into principal components that differentiate speaking styles and speaker groups. In particular, the first principal component captures global pitch elevation patterns, separating adult from child speech and distinguishing male from female speakers, while secondary components encode contour tilt and finer intonational shape differences. Their key result was that when dynamic F0 reconstruction is combined with systematic formant shifts (F1–F3), de-identification performance improves significantly, yielding relative EER increases of up to 25% compared to formant-only anonymization and reinforcing the view that speaker identity is also "hidden" in the temporal pitch trajectories.

3.3.2 Phoneme Durations and Phone Diversity

The importance of analyzing the phoneme durations as a potential privacy leakage channel is demonstrated through a number of papers in speaker verification systems. As early as in the work of ?, speaker-specific segmental duration per phone are put into the spotlight as an underinvestigated factor of verifying speaker identity. In their method the authors utilize a verification system based on phone durations that relies on forced phonetic alignment and computation of average durations for each phone class. Their results show that segmental durational characteristics alone provide meaningful discriminative power and that certain phones exhibit particularly strong speaker-discriminative ability. More specifically, vowels in stressed and pre-stressed positions, as well as sonorant consonants, are found to carry especially high identity information, with sex-specific variations. In more recent years, ? design an attacker that relies exclusively on phoneme duration statistics, and show that

anonymization systems that preserve phoneme timing patterns remain vulnerable in terms of privacy. In more detail, they compute speaker-specific mean duration vectors and apply basic distance metrics (cosine) which indicate that speakers can be re-identified with extremely low EERs on original speech and on anonymization systems not using phoneme duration modifications, like B3 on the VPC of 2024. On the other hand, when an anonymization system regenerates speech and alters durations, it performs much higher in terms of EER, showcasing a better privacy preservation. In a later adaptation ? move to a more advanced neural attacker that learns context dependent duration embeddings. Tested on B3 of VPC 2024 (which modifies utterance pitch and energy), this new approach elevates the ASV performance achieving EERs as low as 1.8%. This approach captures along the individual speaking rate, rhythmic patterns conditioned on the phonetic context, as well as the sequential structure of the phoneme. Complementary evidence that phoneme durations encode speaker information is provided by ? and even though their work focuses on multi-speaker speech synthesis rather than voice privacy, they derive speaker embeddings solely from phoneme identities and their durations and use them as input to a speaker identification model, to check if it would be able to remap to the original speakers. Their experiments achieve a moderate speaker discrimination performance (15.2% EER), indicating once again that phoneme timing patterns reflect partially the original speaker identity.

Noteworthy in understanding how an attacker re-identifies a speaker is also the contribution of individual phonetic variation of the individual. ? in their investigation on how prosodic and phonetic factors contribute to identity leakage in Voice Conversion systems (VC), focus on manipulating phone durations and phonetic variation. From their results it is acquired that when phone durations are progressively modified in the output speech, EER increases once again confirming that duration patterns encode speaker-specific information. However, it is proved that by reducing phonetic variation (in terms of adopting fewer clusters to quantize the target features of each phone) stronger privacy gains are achieved.

A more recent study by ? demonstrates that speaker identity can be partially inferred from speaker-specific distributions of phones and vocabulary, rather than

purely from phone duration patterns. This finding highlights the need to disentangle temporal dynamics as biometric cues from linguistic content as a confounding factor in privacy evaluation because in real-life scenarios, speakers are unlikely to share uniform contextual content across their utterances, and their speech is typically spontaneous and characterized by greater linguistic variability. Lastly, also for phonetic diversity, ? define phonetic richness as the count of distinct phonemes present in an utterance and show that it correlates positively with ASV scores, an effect that is especially seen for utterances with repetitive content. Their results indicate that not only duration but also the sheer inventory of phonemes realized in an utterance carries speaker-discriminative information.

3.3.3 Rhythm

Extending the investigation on rhythm-based speaker representations, ? propose a transformer-based speaker identification model trained solely on frame-aligned character sequences derived from ASR, no energy features, no pitch and no spectral features. Evaluated on LibriSpeech and VoxCeleb1, their rhythm only system achieves an above "chance" accuracy on read speech (0.39 on LibriSpeech), but performs markedly worse on spontaneous speech (0.032 on VoxCeleb1). These findings suggest that rhythmic structure does constitute an identifiable speaker-specific cue and its discrimination is influenced by speech context and intra-speaker variability. From a privacy perspective, this indicates that rhythm-based identity leakage may be dependent on the corpus and therefore suggests a strong re-identification cue for the adversary in the VPC 2024. There has also been extensive research on how linguistic rhythm can be captured through simple duration based statistics derived from the segmentation of speech in consonants and vowels and how this information can be speaker-specific (?). Specifically, linguists have utilized measures such as the **Proportion of Vocalic Intervals (%V)**, the **Standard Deviation of Vocalic Intervals (ΔV)** and also the **Standard Deviation of Consonantal Intervals (ΔC)**, demonstrating that these features reliably distinguish between rhythm classes and suggesting that rhythm is encoded in the distributional properties of segmental timing. Under this theory similar rhythmic properties may encode

speaker-specific "signatures" that persist after anonymization, thereby constituting a potential privacy leakage channel. Another metric that looks like it is able to distinguish differences in stress- and syllable-timing is again based on vowels and is the **Pairwise Variability Index (PVI)** (?). This metric captures the normalized durational variability between successive vowels in a sentence and has been shown to reliably distinguish rhythmic typologies. Although originally applied at the language level, it could theoretically provide a grounded way of quantifying local temporal alternation patterns in speaker anonymization and offer an additional testing cue on whether speakers exhibit stable rhythmic signatures that may persist after anonymization. At last, in terms of consonants another proposed rhythm metric is the **Variation Coefficient of Consonantal Intervals (varco ΔC)**, defined as the standard deviation of consonantal interval durations divided by their mean duration and scaled by 100 (?). By normalizing ΔC with respect to the mean consonantal duration, varco ΔC captures relative variability and disentangles rhythmic structure from global tempo effects. In the context of voice anonymization, this rate normalized metric could allow for an assessment of whether attackers exploit rhythmic structure or just primarily global temporal characteristics.

3.4 Correlation of Prosodic and Linguistic Features

Another hypothesis is that the combination of *what* a speaker says and *how* it is said may jointly constitute a cue exploitable by an attacker model for re-identification. Supporting this from a syntactic and semantic perspective, ? investigate the extent to which deep speech encoders capture grammatical distinctions directly from acoustic input. Using a minimal pair probing framework, they construct controlled speech stimuli differing in specific syntactic properties and apply layer-wise linear probes to representations from multiple pretrained speech models. Their experiments show that syntactic distinctions are linearly recoverable from acoustic representations across several architectures and layers and that time-resolved probing reveals grammatical information becoming detectable early in the acoustic signal. Such evidence supports the hypothesis that temporal dynamics and other acoustic cues may

correlate with syntactic and lexical choices and form identity cues when combined with speaker-specific phonetic patterns.

4 Methodology

This chapter describes the methodological pipeline used to evaluate speaker anonymization performance and investigate speaker-specific vulnerability after anonymization. The methodology consists of three main parts: Reproduction of the VPC 2024 baseline anonymization systems and computation of global and speaker-level EER, evaluation using the ZEBRA framework, Linkability, Singling Out, and a feature-level analysis examining whether speaker vulnerability can be linked to preserved phonetic, acoustic, or linguistic information.

4.1 Experimental Setup: Reproduction of the VPC 2024 Baseline Systems

Three official VPC 2024 baseline systems are reproduced: B3, B4, and B5, representing fundamentally different anonymization paradigms. The official VPC 2024 repository² is used as the codebase throughout this thesis, with the official configuration files (`configs/anon_sttts.yaml`, `configs/anon_nac.yaml`, and `configs/anon_asrbn.yaml` for B3, B4, and B5 respectively) modified as needed when necessary. The semi-informed attacker adopted for this challenge, is trained on the anonymized versions of LibriSpeech `train-clean-360` (?) produced by each baseline as proposed by the protocol. Then, anonymization is applied at utterance level to the VPC 2024 LibriSpeech development and test partitions (four partitions in total: `dev-f`, `dev-m`, `test-f`, `test-m`).

²<https://github.com/Voice-Privacy-Challenge/Voice-Privacy-Challenge-2024>

4.1.1 B3: Phonetic Transcriptions and GAN-based Speaker Anonymization

Baseline B3 (from now on: *STTTS*) is a speech synthesis based system that preserves linguistic and prosodic information while replacing the speaker embedding. It extracts: A speaker embedding, phonetic transcription, $F0$, energy, and phone durations from the input waveform. Then, the speaker embedding gets replaced by a pseudospeaker embedding generated from a Wasserstein GAN, accepted only if sufficiently dissimilar from the original via a cosine-distance criterion. Pitch and energy are randomly perturbed at the phone level. As a last step, the anonymized waveform is synthesized using a FastSpeech2 acoustic model and a HiFi-GAN vocoder (?).

4.1.2 B4: Neural Audio Codec Language Modeling

Baseline B4 (from now on: *NAC*) treats anonymization as a sequence generation problem in token space. Semantic tokens representing the linguistic content of the source utterance are concatenated with acoustic tokens sampled from a pool of pseudospeakers, and a decoder transformer generates new acoustic tokens satisfying both semantic consistency and constraints of the pseudospeaker’s style. The generated tokens are then decoded back into a waveform via a neural audio codec decoder (?).

4.1.3 B5: ASR Bottleneck Features with Vector Quantization

Baseline B5 (from now on: *ASRBN*) reduces speaker ID leakage at the feature level through vector quantization of ASR bottleneck features. The system combines a pretrained wav2vec2 model with three TDNN-F layers, applying vector quantization to the inner bottleneck representation of the final layer in order to suppress speaker information. The resulting VQ-BN features, pitch, and a one-hot pseudospeaker vector are passed to a HiFi-GAN network for the last step of waveform synthesis.

4.1.4 ASV Evaluation with Global EER and Speaker-Level EER

For each of the three reproduced baselines, the official VPC 2024 ASV evaluation pipeline is applied, by performing the needed code and configuration changes, in order to exclude the utility evaluation. Under the semi-informed attacker scenario, the ECAPA-TDNN attacker is trained for each anonymization condition using the corresponding anonymized version of LibriSpeech `train-clean-360`, letting the attacker adjust to the same embedding space as the anonymized speech. The official global EER is then computed separately for each condition over the LibriSpeech enrollment-trial pairs (`dev-f`, `dev-m`, `test-f`, `test-m`), as the pipeline suggests.

The first evaluation modification comes from the acquisition of speaker-level EER scores by adapting the official ASV evaluation code, similarly to ?. After trial-level scores are computed, the score table is grouped by enrollment speaker identifier, and for each speaker s with available target scores $\mathcal{T}_s^+$ and non-target scores $\mathcal{T}_s^-$. Speakers for whom only one score type is available are excluded. The reason for this choice is that in EER calculation, the miss rate computation needs target trials, while the false alarm rate needs the non-target trials.

Therefore, the speaker-level EER gets defined as:

$$(9) \quad \text{EER}_s = \text{EER}(\mathcal{T}_s^+, \mathcal{T}_s^-).$$

The global EER produced by the ASV pipeline is kept unchanged, with the speaker-level analysis serving as an additional diagnostic tool for getting an insight of the individual’s behaviour rather than a suggested replacement.

4.2 Alternative Metrics Evaluation

4.2.1 ZEBRA Framework

The first set of metrics that are compared to EER, is taken from the ZEBRA framework (?), which evaluates the privacy disclosure from an evidence-based perspective using Empirical Cross Entropy (ECE) profiles and calibrated log-likelihood ratios

(LLRs), unlike EER which identifies a single threshold. The framework includes two complementary measures. On the one hand, **dECE** measures the difference between the ECE of a system with zero evidence (meaning perfect privacy) and the ECE profile of the evaluated ASV system’s scores over the adversary’s prior beliefs. A value of 0 in this metric indicates that the ASV scores (which in the current setup correspond to cosine-based x-vector scoring) provide no identity evidence beyond the adversary’s prior, after it has "heard" the anonymized speech, while higher values indicate residual speaker-discriminative information. On the other hand, **max_abs_LL**R captures the strongest individual evidence observed after ASV score-to-LLR calibration, serving as a worst-case disclosure measure. The notion here is that an anonymization system may provide acceptable average population protection while still leaving individual speakers highly exposed.

The core ZEBRA computations rely on the existing ZEBRA functions in the VPC evaluation framework for privacy assessment. The additional implementation work consists of adapting the acquired VPC ASV score outputs to the input format required by the ZEBRA framework. In more detail, the original VPC score files contain four columns:

$$(10) \quad [\text{enroll_id}, \text{trial_id}, \text{label}, \text{score}].$$

The added preprocessing step generates the two framework-required ZEBRA input files: a **scores_zebra** file retaining enrollment ID, trial ID, and score:

$$(11) \quad [\text{enroll_id}, \text{trial_id}, \text{score}],$$

and a **key_zebra** file converting the ASV binary labels to textual trial types:

$$(12) \quad 1 \rightarrow \text{target}, \quad 0 \rightarrow \text{nontarget},$$

yielding:

$$(13) \quad [\text{enroll_id}, \text{trial_id}, \text{trial_type}].$$

Since the ASV scores are raw cosine similarity scores rather than calibrated LLR values (which the framework accepts), they have to be transformed using the Pool

Adjacent Violators (PAV) calibration, which estimates a monotonic mapping from raw scores to posterior probabilities and converts them into LLRs by removing the empirical prior log-odds. Laplace smoothing is then applied to avoid the infinite LLR values at the score distribution extremes. Letting $\mathcal{S}^+$ and $\mathcal{S}^-$ denote target and non-target scores respectively, PAV calibration produces the calibrated LLRs associated with the target ($\mathcal{L}^+$) and non-target ($\mathcal{L}^-$) scores:

$$(14) \quad \mathcal{L}^+, \mathcal{L}^- = \text{Calibrate}_{\text{PAV}}(\mathcal{S}^+, \mathcal{S}^-).$$

The global ZEBRA metrics are then computed as:

$$(15) \quad \text{dECE} = \frac{\text{ECE}(\mathcal{L}^+) + \text{ECE}(-\mathcal{L}^-)}{\log(2)},$$

$$(16) \quad \text{max_abs_LLR} = \frac{\max(|\mathcal{L}^+ \cup \mathcal{L}^-|)}{\log(10)},$$

where dividing by $\log(10)$ expresses the worst-case evidence in base 10 LLR units. The `max_abs_LL` value is mapped to an evidence tag from A to F, with lower tags indicating weaker evidence and stronger privacy, in the same way the framework defines. At the end, for each evaluated condition, the implementation outputs:

- A calibrated LLR score file,
- A global ZEBRA summary,
- An ECE plot over the attacker’s priors comparing the system curve against the zero-evidence reference (as seen in the Appendix ??).

As an additional diagnostic analysis, the ZEBRA computation is extended to the speaker-level to examine whether speakers with low EER also show higher evidence leakage across the majority of their anonymized utterances (`dECE`). For each enrollment speaker s , trials are extracted and divided into target and non-target subsets, excluding speakers for whom only one trial type was available. This filtering step is necessary because `dECE` is computed from calibrated LLRs estimated from both

target and non-target score distributions. PAV calibration is then applied locally per speaker:

$$(17) \quad \mathcal{L}_s^+, \mathcal{L}_s^- = \text{Calibrate}_{\text{PAV}}(\mathcal{S}_s^+, \mathcal{S}_s^-).$$

The speaker-level dECE is computed as:

$$(18) \quad \text{dECE}_s = \frac{\text{ECE}(\mathcal{L}_s^+) + \text{ECE}(-\mathcal{L}_s^-)}{\log(2)},$$

and the speaker-level worst-case evidence as:

$$(19) \quad \text{max_abs_LLR}_s = \frac{\max(|\mathcal{L}_s^+ \cup \mathcal{L}_s^-|)}{\log(10)}.$$

This pipeline produces a per-speaker ZEBRA table for every trial partition of the form:

$$(20) \quad [\text{speaker}, \text{n_trials}, \text{n_tar}, \text{n_non}, \text{dECE}, \text{max_abs_LLR}, \text{tag}],$$

where `speaker` denotes the enrollment speaker ID, `n_trials` is the total number of trials for that speaker, `n_tar` and `n_non` denote the number of target and non-target trials respectively, `dECE` measures the speaker-level deviation from the zero-evidence privacy condition, and `max_abs_LL`R captures the strongest calibrated evidence observed among that speaker’s trials. The `tag` field provides the leakage A-F ranking based on `max_abs_LL`R.

4.2.2 Linkability & Singling Out

The next two metrics are the legally inspired Linkability & Singling Out proposed by ?. First, Linkability measures whether an attacker can correctly associate a trial embedding with the corresponding enrollment speaker among the trial candidates, while Singling Out measures whether the attacker can *isolate* exactly one individual from the set of anonymized speech samples. In a different manner from the EER and ZEBRA initializations, Linkability & Singling Out are computed using directly the speaker embeddings, rather than the ASV scores.

These metrics are implemented based on the official repository of the paper ³, but adapted to the VPC 2024 setup using the x-vector embeddings already produced from the anonymization pipelines. Unlike the original paper, which uses Common Voice A and B as disjoint subsets, the current experiments use the LibriSpeech enrollment and trial partitions already met in the VPC 2024 pipeline. The needed enrollment embeddings are loaded from the pooled development and test enrollment directories, namely (`libri_dev_enrolls` and `libri_test_enrolls`), and the trial embeddings are loaded from the female and male development and test trial directories per anonymization method. For this experiment, pooling across the development and test sets is necessary in order to obtain a sufficient number of speakers and utterances for evaluation across multiple population sizes and conversation lengths. This setup adaptation follows the logic of the original Linkability & Singling Out framework, where enrollment and test material are disjoint at the utterance level but contain overlapping speakers. Thus, the same-speaker identities are used to define the enrollment references and the trial population, but the actual enrollment and trial embeddings are derived from separate utterance partitions, preserving the required separation between enrollment and trial material. Lastly, speaker representations are acquired from `speaker_vectors.pt` files per anonymization method. There, the enrollment data are represented by a single speaker-level embedding and trial data are grouped by speaker using `id2idx` and `idx2spk` mappings.

Linkage is successful when the cosine similarity between the test embedding and the correct enrollment embedding exceeds that of all the other candidates:

$$(21) \quad s(x_i^{\text{test}}, x_i^{\text{enroll}}) > \max_{j \neq i} s(x_i^{\text{test}}, x_j^{\text{enroll}}).$$

With Linkability defined as the probability of a successful linkage:

$$(22) \quad \pi_{\text{link}} = \Pr_{x_i^{\text{test}}} \left[s(x_i^{\text{test}}, x_i^{\text{enroll}}) > \max_{j \neq i} s(x_i^{\text{test}}, x_j^{\text{enroll}}) \right].$$

The evaluation is conducted across the following enrollment population sizes:

$$(23) \quad N' \in \{5, 10, 15, 20, 25, 30, 40, 50\},$$

³<https://github.com/brijmohan/voice-anonymization-legal-eval>

and the following conversation lengths:

$$(24) \quad L \in \{1, 3, 10, 30\}.$$

The smaller N' range compared to the original paper (which numbers up to 22,024) reflects the smaller number of common speakers available in the pooled LibriSpeech trial sets. Moreover, $L = 10$ is added as an intermediate setting beyond the original paper’s $\{1, 3, 30\}$ to observe the transition between short and long conversation conditions more gradually. For each test speaker, L utterance-level embeddings are sampled and averaged into a single test embedding and speakers who have fewer than L utterances are excluded because the L condition must be constructed from the same number of utterances for every evaluated speaker. For each value of N' , a candidate enrollment set is constructed from the correct speaker and $N' - 1$ randomly sampled impostors, like suggested in the paper. Finally, Linkability is estimated as:

$$(25) \quad \hat{\pi}_{\text{link}} = \frac{1}{|\mathcal{T}_L|} \sum_{i \in \mathcal{T}_L} \mathbf{1} \left[\arg \max_{j \in \mathcal{C}_i} s(x_i^{\text{test}}, x_j^{\text{enroll}}) = i \right],$$

where $\mathcal{T}_L$ is the set of test speakers with at least L utterances and $\mathcal{C}_i$ the candidate enrollment set for speaker i . The experiment is repeated over 20 random runs per (L, N') pair (more than the original paper’s 5 runs) to obtain more stable estimates given the much smaller data samples.

For Singling Out the attacker defines a binary predicate that fires when similarity to an enrollment embedding exceeds the calibrated threshold:

$$(26) \quad p(x^{\text{test}}) = \mathbf{1} \left[s(x^{\text{test}}, x^{\text{enroll}}) > s^{\text{thresh}} \right],$$

with Singling Out succeeding if exactly one test speaker satisfies the calibrated predicate:

$$(27) \quad \sum_{i=1}^N p(x_i^{\text{test}}) = 1.$$

The random predicate baseline is $\exp(-1) \approx 0.3679$. Evaluation here is conducted over $N \in \{20, 30, 40, 50\}$ and $L \in \{1, 3, 10, 30\}$. The smaller N range compared to

the original paper (which numbers up to 22,024) again reflects the limited number of eligible speakers in the VPC setup. For each speaker, utterance embeddings are permuted and split into disjoint groups of size L , averaged into grouped embeddings, with up to 10 groups per speaker. Only speakers with at least three groups are retained (this design choice is stricter than the original paper’s minimum of $K \geq 2$) in order to ensure calibration stability, guaranteeing at least two calibration groups and one test group per speaker.

To continue, the evaluation follows a cross-validation procedure: In each fold, one grouped embedding serves as the test embedding and the remainder as calibration. With K grouped embeddings per speaker, the number of calibration groups is $M = K - 1$. The threshold is set as the average of the M -th and $(M + 1)$ -th highest pooled calibration score:

$$(28) \quad s^{\text{thresh}} = \frac{1}{2} (s_{(M)} + s_{(M+1)}),$$

generalizing the original paper’s fixed $M = 9$ rule to variable data availability. The predicate is then applied to the N test embeddings, recording success if exactly one exceeds the predicate/threshold:

$$(29) \quad \sum_{i=1}^N \mathbf{1} [s(x_i^{\text{test}}, x^{\text{enroll}}) > s^{\text{thresh}}] = 1.$$

For each random run, N speaker identities are sampled from the set of speakers that are present in both the pooled enrollment and trial partitions. The enrollment embedding used to define the predicate is loaded from the enrollment set, while the calibration and test embeddings are constructed from the corresponding anonymized trial utterances. The full experiment is repeated over five random runs for each (L, N) combination, following the sampling strategy of the original framework.

4.3 Phonetic Features Analysis

Next, a feature-level analysis examines whether speakers who remain more easily re-identifiable (based on ASV similarity and individual EER scores) preserve specific phonetic/acoustic patterns in their anonymized speech. The motivation is that

speaker identity may be reflected in interpretable speech properties (e.g. phoneme durations, rhythm, fundamental frequency, formant trajectories).

Initially, phone-level alignments are generated for all trial data (original and anonymized) using the Montreal Forced Aligner (MFA) (?). MFA takes as input the speech waveform and its corresponding transcription and produces a time-aligned TextGrid file containing phone-level intervals with start and end times, from which the phone durations per utterance are extracted. The alignments are applied consistently across all three anonymization systems and both genders on the four trial partitions, preserving original utterance identifiers so that the extracted phone intervals can be matched with the speaker-level ASV results.

The analysis is performed at utterance level and averaged to speaker level. Let x_u^{orig} denote the original utterance and x_u^{anon} the corresponding anonymized utterance for utterance u . For each pair, phonetic, prosodic, acoustic, and embedding-based representations are extracted and compared either directly between x_u^{orig} and x_u^{anon} , or indirectly by comparing anonymized utterances against their original speaker centroids.

4.3.1 Rhythm Features

The first feature group consists of rhythm metrics derived from the precalculated MFA alignments. For this and the following analyses, silence and non-speech labels are excluded, and each remaining phone gets assigned to a vowel or consonant class. For each utterance, the following rhythm features get computed separately for both the original and their corresponding anonymized utterances:

- Percentage of vocalic intervals (?),
- Percentage of consonantal intervals (?),
- Mean vowel duration,
- Mean consonant duration,

- Standard deviation of vowel and consonant durations (ΔV and ΔC) (?),
- Varco-V and Varco-C, i.e. variation coefficients that normalize ΔV and ΔC by the corresponding mean vowel and consonant durations (?),
- Raw and normalized pairwise variability index for vowels (rPVI-V and nPVI-V) (?),
- Raw and normalized pairwise variability index for consonants (rPVI-C and nPVI-C) (?).

As a next step, two additional comparison measures are derived for each rhythm feature. These values are not independent rhythm metrics, but change indicators that describe how anonymization alters each rhythmic property. So, for each utterance u and rhythm feature f , the first value, the **signed shift**, is defined as:

$$(30) \quad s_{u,f} = f(x_u^{anon}) - f(x_u^{orig}),$$

while the corresponding **absolute shift** is defined as:

$$(31) \quad a_{u,f} = |f(x_u^{anon}) - f(x_u^{orig})|.$$

The signed shift captures the direction of the change caused by anonymization. If positive it means that the rhythm feature is higher than in the original utterance, while a negative value means that the feature is "more suppressed" after anonymization. The absolute shift captures the magnitude of the change, meaning how strongly a rhythmic feature is modified by anonymization, regardless of whether the feature's values increase or decrease.

All of these features are then averaged at the speaker level in order to summarize how much, and in which direction, each speaker's rhythmic profile is modified after anonymization.

4.3.2 Phone-Duration Features

The second feature group focuses on checking if phone durations are preserved in the anonymized speech. Based on the MFA alignments, original and anonymized phone

intervals are compared by position. For each phone i in utterance u , the **duration shift** and **duration ratio** are defined as:

$$(32) \quad d_{u,i}^{\text{shift}} = d_{u,i}^{\text{anon}} - d_{u,i}^{\text{orig}}, \quad r_{u,i} = \frac{d_{u,i}^{\text{anon}}}{d_{u,i}^{\text{orig}}},$$

capturing the **absolute** and **relative** changes in segmental timing respectively. At the utterance level, the analysis computes individual phone counts, total and mean phone duration, phone rate and phone rate shift, as well as Pearson and Spearman correlations between original and anonymized phone durations by position. Additionally, local transition features are extracted for $V \rightarrow V$, $V \rightarrow C$, $C \rightarrow V$, and $C \rightarrow C$ in order to examine whether local timing relations between consecutive phones were preserved after anonymization. Finally, speaker-level summaries of these relative measures are obtained by averaging the utterance-level measures per speaker.

4.3.3 Acoustic and Prosodic Features

The third feature group examines preservation of acoustic and prosodic characteristics between original and anonymized utterances.

The acoustic descriptors get extracted using openSMILE (?) through the eGeMAPS (?) feature set, covering among others, F0, formant frequencies and bandwidths, jitter, shimmer, MFCCs, loudness (energy). For each, speaker preservation is summarized via Pearson and Spearman correlations and mean signed and absolute shifts between the original and the anonymized feature values.

Pitch and formant trajectory preservation is analyzed using Parselmouth (?), with F0 getting extracted at a 10 ms time step and F1, F2, F3 getting estimated via Praat (?) formant analysis. For each utterance u , the original trajectory z_u^{orig} and anonymized trajectory z_u^{anon} are compared using Pearson and Spearman correlations and Dynamic Time Warping distance (DTW) (?), which allows local temporal realignment before computing distance and is therefore suitable for cases where anonymization slightly stretches or compresses utterance timing while preserving

the overall contour shape. The correlation-based preservation score is averaged per speaker and defined as:

$$(33) \quad \rho_u = \text{corr}(z_u^{\text{orig}}, z_u^{\text{anon}}).$$

4.3.4 ASV Embedding Similarity

The fourth analysis step uses the ASV embedding similarity of the original and anonymized x-vector embeddings, loaded from the ASV evaluation outputs. For each utterance, the cosine similarity between the original and anonymized embedding is computed as:

$$(34) \quad \text{cos}(e_u^{\text{orig}}, e_u^{\text{anon}}) = \frac{e_u^{\text{orig}} \cdot e_u^{\text{anon}}}{\|e_u^{\text{orig}}\| \|e_u^{\text{anon}}\|}.$$

The speaker-level embedding centroids are also computed by averaging utterance-level embeddings for each speaker. Then the similarity between the original centroid of a speaker and the anonymized centroid of the same speaker gets calculated, as well as the similarity to the anonymized centroids of other speakers. The motivation of this is to estimate whether the anonymized representation of a speaker remains closer to its own original representation than to other speakers.

4.3.5 Linking Phonetic Features with ASV Scores

For the feature-space similarity analysis (as reported in Subsection ??, Table ?? and Table ??), the individual features within each of the extracted feature groups get combined into a multi-dimensional feature vector for each utterance. These vectors are standardized within each feature space, and original speaker centroids are computed by averaging the original utterance-level vectors for each speaker. Cosine similarity is then measured between each anonymized trial-utterance vector and

the corresponding original enrollment-speaker centroid. Target trials correspond to same-speaker trials, while non-target trials correspond to different-speaker trials.

The target minus non-target feature-space separation is then computed as:

$$(35) \quad \Delta_F = \mu(S_F^{\text{tar}}) - \mu(S_F^{\text{non}}),$$

where S_F^{tar} and S_F^{non} are the target and non-target cosine similarity distributions for feature space F . Larger positive values indicate that same-speaker trials remain more similar than different-speaker trials after anonymization.

4.3.6 Correlation between Phonetic Features and EER Scores

The speaker-level relationship between a phonetic preservation feature (i.e. rhythm, F0, formant trajectory, and phone-duration) and EER is estimated as:

$$(36) \quad r = \text{corr}(F_s, \text{EER}_s).$$

A negative correlation indicates that stronger preservation of a feature is associated with lower EER and therefore higher re-identification vulnerability, while a positive correlation indicates association with higher EER and stronger privacy. The preservation features reflect similarity between original and anonymized speech while the shift and distance features (signed shifts, absolute shifts, phone-duration shifts, phone-rate shifts, DTW distance) reflect the amount of change introduced by anonymization. For the cause of this thesis, which is to explore if preservation of phonetic/acoustic features helps the attacker re-identify an individual, only the results of the negative correlations are reported (as seen in Table ??).

4.3.7 Linguistic and Acoustic Interaction Features

At last, a linguistic interaction analysis is conducted to examine whether lexical similarity between enrollment and trial material contributes to residual speaker recognizability after anonymization, independently and in combination with phonetic and acoustic features.

Trial and enrollment transcripts are initially normalized, then for each enrollment speaker, all enrollment transcripts are concatenated into a speaker-level text profile, and a TF-IDF vectorizer using word unigrams and bigrams is fitted on both the enrollment profiles and trial transcripts. Lexical similarity for each ASV trial pair is then computed as:

$$(37) \quad \text{sim}_{\text{text}}(e, u) = \cos(v_u^{\text{text}}, v_e^{\text{text}}),$$

where v_u^{text} and v_e^{text} are the TF-IDF representations of trial utterance u and enrollment speaker e respectively. Additional lexical statistics are computed per trial utterance, namely: Word count, character count, type-token ratio, function-word ratio, and average word length.

Then, the lexical features are merged with the phonetic and acoustic feature space similarities computed earlier (rhythm, openSMILE, F0 preservation, trajectory preservation, phone-duration preservation, and ASV embedding similarity) and with the ASV score table. The interaction features are created by multiplying text similarity with each phonetic/acoustic similarity feature in the following manner:

$$(38) \quad I_{u,k} = \text{sim}_{\text{text}}(e, u) \cdot \text{sim}_k(e, u),$$

where $\text{sim}_k(e, u)$ denotes the similarity value from phonetic or acoustic feature space k . **Interaction** tests whether ASV scores are better explained when lexical and phonetic/acoustic similarity are considered jointly.

The merged table is analyzed using Pearson and Spearman correlations between ASV scores and all feature groups, followed by Ridge regression to compare the predictive power of four feature configurations: Text-only, phonetic/acoustic-only, combined, and combined with interaction terms. Ridge Regression is selected due to the correlated variables and interaction terms in the feature sets, with the ridge penalty providing more robust coefficient estimates. Model performance is evaluated using five-fold shuffled cross-validation at the ASV trial-pair level, with a fixed random seed. The folds are not speaker-grouped, so the reported cross-validated R^2

may be optimistic for unseen-speaker generalization:

$$(39) \quad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2},$$

where y_i is the true ASV score, $\hat{y}_i$ the predicted score, and $\bar{y}$ the mean true score. Additionally, Logistic Regression is applied to test whether the same feature groups could distinguish target from non-target trials, evaluated using cross-validated ROC-AUC, interpreted as the probability that a randomly selected target trial receives a higher predicted score than a randomly selected non-target trial, with 0.5 corresponding to chance and values closer to 1.0 indicating stronger separation. Together, R^2 and ROC-AUC address two complementary questions:

- How much of the variation in ASV scores can be explained by each feature group (namely: Text-only features, phonetic/acoustic-only features, combined features, and interaction-based features)? (as demonstrated in Table ??)
- Do these feature groups preserve enough speaker-discriminative information to separate target trials from non-target trials, as reflected by the ROC-AUC score? (as demonstrated in Table ??)

5 Results

In the current chapter, the results of the experiments and analyses described in Section ?? are presented. First, the results from the trial partitions of LibriSpeech in VPC 2024, namely `dev-f`, `dev-m`, `test-f`, and `test-m`, are reported for global EER, global dECE according to the ZEBRA framework, speaker-level EER, and speaker-level dECE. Next, the results for Linkability & Singling Out are presented and lastly, the results of the phonetic, acoustic, embedding-based, and linguistic feature analyses are reported. The analysis is presented through feature-space target/non-target separation, correlations with ASV similarity scores, predictive modeling of ASV-score variation, and speaker-level correlations with individual EER scores.

5.1 Results for EER & ZEBRA

Table ?? reports per-speaker ZEBRA, dECE and EER results for the LibriSpeech development female trial set. As seen, the ASRBN pipeline achieves the best global privacy performance with the lowest dECE (0.073) and the highest EER (34.231%). The NAC system follows with a dECE of 0.088 and an EER of 33.949%, while the STTTS baseline performs the worst with a dECE of 0.136 and an EER of 28.265%.

dev_f												
Spk.	B3 dECE	DR	B3 EER	ER	B4 dECE	DR	B4 EER	ER	B5 dECE	DR	B5 EER	ER
84	0.253	8	22.873	8	0.000	1	61.105	1	0.005	1	69.142	1
1462	0.343	11	18.492	11	0.143	9	31.718	6	0.194	12	22.708	13
1673	0.452	15	10.078	15	0.388	15	18.651	15	0.412	14	14.515	14
1988	0.020	3	46.715	3	0.008	2	55.476	2	0.032	2	48.853	2
1993	0.378	12	14.824	13	0.072	4	34.073	5	0.047	4	37.035	7
2035	0.202	6	29.979	5	0.141	8	29.979	9	0.050	6	26.498	12
2277	0.210	7	27.349	7	0.084	5	38.332	4	0.149	10	30.148	9
2412	0.401	14	17.083	12	0.243	13	29.271	10	0.491	15	12.188	15
3536	0.286	9	21.215	10	0.201	11	30.278	8	0.094	8	39.342	4
5338	0.121	4	36.354	4	0.124	7	31.412	7	0.034	3	39.082	6
5895	0.311	10	21.988	9	0.168	10	29.134	11	0.048	5	39.463	3
6313	0.386	13	14.721	14	0.087	6	19.949	13	0.166	11	29.391	10
6319	0.176	5	29.391	6	0.318	14	19.949	13	0.196	13	29.391	10
6345	0.016	2	48.535	2	0.052	3	43.258	3	0.075	7	39.289	5
8842	0.004	1	54.738	1	0.217	12	23.790	12	0.135	9	30.845	8
Global	0.136	–	28.265	–	0.088	–	33.949	–	0.073	–	34.231	–

Table 1: Per-speaker ZEBRA dECE, EER, and performance ranks for individual speakers for B3, B4, B5 on dev-f, with ranking of 1 meaning best privacy preservation. DR=dECE speaker rank, ER=EER speaker rank. Red: Speakers with EER below 30% across all systems.

At the speaker level though, substantial variability between speakers and systems can be observed. More specifically, speaker **84** seems strongly protected under NAC and ASRBN (EERs of 61.105% and 69.142% and dECE of 0 in NAC), while speaker **1673** remains very vulnerable across all systems with constant EERs under 20% and also ranking last under dECE across baselines. Another interesting finding is that EER and dECE do not always agree in speaker rankings: Specifically under

NAC, speaker **6313** has a low EER of 19.949% (suggesting big privacy risk), but a low dECE of 0.087 suggesting otherwise. Speaker **3536** has an EER of 30.278% suggesting protection but a higher dECE of 0.201. Lastly, speaker **2412** shows a similar mismatch with an EER close to 30% but a moderate dECE of 0.243.

Table ?? shows that NAC and ASRBN perform similarly on dev-m, with a dECE of 0.118 for both and an EER of 30.588% and 30.740% respectively. STTTS performs worse with a dECE of 0.196 and an EER of 23.788%. Speakers **652**, **2803**, **2902**, **5536**, **7976**, and **8297** remain vulnerable across all three systems making this trial set the one with the most vulnerable individuals. Speaker **2803** is particularly exposed, with EER values of 9.174%, 18.946%, and 17.102% and quite high dECE across all systems. Again, an EER and dECE diverge is noted for some speakers: For example, under NAC, speaker **6241** has a dECE of 0.145 indicating leakage while their EER of 31.659% suggests adequate protection.

dev_m												
Spk.	B3 dECE	DR	B3 EER	ER	B4 dECE	DR	B4 EER	ER	B5 dECE	DR	B5 EER	ER
174	0.122	1	36.129	1	0.163	8	31.241	7	0.150	7	30.538	7
251	0.124	2	33.333	3	0.118	6	31.528	6	0.105	5	37.274	4
652	0.581	14	8.011	13	0.174	9	25.802	9	0.406	14	10.000	14
1272	0.167	7	25.847	10	0.105	5	29.622	8	0.051	3	39.617	2
2428	0.214	9	27.283	6	0.048	2	40.924	3	0.051	2	41.197	1
2803	0.501	12	9.174	12	0.386	14	18.946	13	0.399	13	17.102	13
2902	0.576	13	7.130	14	0.190	10	24.596	10	0.236	10	29.393	8
3752	0.127	3	29.719	5	0.061	3	35.722	4	0.065	4	38.668	3
5536	0.359	11	15.401	11	0.385	13	12.816	14	0.177	9	28.543	10
5694	0.135	4	30.182	4	0.063	4	42.385	2	0.110	6	35.444	6
6241	0.137	5	33.868	2	0.145	7	31.659	5	0.311	12	22.209	12
6295	0.145	6	27.283	6	0.036	1	45.435	1	0.036	1	36.521	5
7976	0.204	8	26.645	8	0.203	11	23.300	11	0.175	8	29.331	9
8297	0.216	10	26.019	9	0.300	12	19.651	12	0.254	11	25.526	11
Global	0.196	–	23.788	–	0.118	–	30.588	–	0.118	–	30.740	–

Table 2: Per-speaker ZEBRA dECE, EER, and performance ranks for individual speakers for B3, B4, B5 on dev-m, with ranking of 1 meaning best privacy preservation. DR=dECE speaker rank, ER=EER speaker rank. Red: Speakers with EER below 30% across all systems.

Table ?? shows ASRBN performing best on `test-f` with dECE of 0.107 and EER of 32.483%, followed by STTTS (dECE of 0.125, EER of 28.682%) and NAC (dECE 0.142, EER 30.109%). Speakers **1995**, **3570**, and **1221** remain insufficiently protected across all systems. Speaker **3570** is particularly vulnerable, with EER values of 6.886% and 6.533% under STTTS and ASRBN and the highest dECE values for both (0.557 and 0.621). Speaker **2961** illustrates EER-dECE divergence: Their EER of 30.407% under NAC suggests borderline protection, while its dECE of 0.274 ranks them among the most exposed speakers.

test_f												
Spk.	B3 dECE	DR	B3 EER	ER	B4 dECE	DR	B4 EER	ER	B5 dECE	DR	B5 EER	ER
121	0.013	3	54.539	2	0.280	13	22.873	13	0.110	7	45.103	1
237	0.331	11	18.700	10	0.144	7	33.357	4	0.062	3	40.469	4
1284	0.292	8	23.527	8	0.078	2	40.028	2	0.060	2	40.028	5
1580	0.239	7	25.622	7	0.138	6	33.357	4	0.164	10	30.161	10
1995	0.394	12	14.368	13	0.206	9	26.173	11	0.323	15	21.914	15
2961	0.295	9	17.767	11	0.274	12	30.407	8	0.229	12	30.407	9
3570	0.557	16	6.886	16	0.621	16	6.533	16	0.264	14	25.905	13
4446	0.000	1	84.766	1	0.237	11	25.066	12	0.258	13	23.420	14
4970	0.402	13	15.505	12	0.095	3	34.353	3	0.059	1	43.884	3
4992	0.304	10	21.374	9	0.290	14	21.087	14	0.063	4	36.309	6
5142	0.004	2	48.685	4	0.132	5	33.357	4	0.148	9	29.596	12
5683	0.147	5	31.896	6	0.129	4	31.896	7	0.140	8	29.814	11
6829	0.454	14	12.518	14	0.173	8	28.592	9	0.174	11	31.152	8
8463	0.019	4	51.078	3	0.209	10	27.658	10	0.086	6	36.135	7
8555	0.171	6	34.050	5	0.052	1	41.399	1	0.064	5	44.825	2
1221	0.519	15	12.288	15	0.370	15	14.046	15	0.458	16	9.341	16
Global	0.125	–	28.682	–	0.142	–	30.109	–	0.107	–	32.483	–

Table 3: Per-speaker ZEBRA dECE, EER, and performance ranks for individual speakers for B3, B4, B5 on `test-f`, with ranking of 1 meaning best privacy preservation. DR=dECE speaker rank, ER=EER speaker rank. Red: Speakers with EER below 30% across all systems.

On Table ?? `test-m`, ASRBN again performs best with a dECE of 0.116 and an EER of 32.770%, followed by NAC (dECE 0.135, EER 29.619%) and STTTS (dECE 0.210, EER 26.278%). Here the individuals **1320**, **4077**, **7021**, and **8224** remain insufficiently protected across all systems, with particularly high dECE values under

STTTS and NAC. Speaker **61** illustrates a further EER-dECE mismatch: Their EER of 30.244% under ASRBN suggests borderline protection, while their dECE of 0.156 indicates moderate leakage, confirming that EER alone can mask remaining speaker information.

test_m												
Spk.	B3 dECE	DR	B3 EER	ER	B4 dECE	DR	B4 EER	ER	B5 dECE	DR	B5 EER	ER
61	0.335	7	21.294	7	0.086	4	37.815	5	0.156	7	30.244	7
260	0.118	2	38.608	2	0.057	2	43.388	1	0.108	3	31.787	6
908	0.392	10	17.216	10	0.115	6	37.860	4	0.365	13	17.761	13
1089	0.243	4	25.034	4	0.169	7	35.020	7	0.243	9	19.973	11
1320	0.245	5	24.470	5	0.382	11	14.668	11	0.202	8	26.244	9
2830	0.032	1	44.397	1	0.201	9	23.972	9	0.117	6	35.177	3
4077	0.465	12	11.788	12	0.512	12	11.788	12	0.338	11	23.510	10
5105	0.409	11	12.842	11	0.108	5	35.929	6	0.037	2	45.691	2
6930	0.547	13	8.149	13	0.077	3	38.462	3	0.109	4	34.668	5
7021	0.388	9	18.632	9	0.375	10	18.356	10	0.347	12	18.908	12
7127	0.268	6	23.665	6	0.186	8	33.885	8	0.111	5	34.977	4
7176	0.196	3	25.614	3	0.035	1	41.052	2	0.002	1	51.367	1
8224	0.382	8	20.607	8	0.564	13	7.114	13	0.272	10	27.053	8
Global	0.210	–	26.278	–	0.135	–	29.619	–	0.116	–	32.770	–

Table 4: Per-speaker ZEBRA dECE, EER, and performance ranks for individual speakers for B3, B4, B5 on **test-m**, with ranking of 1 meaning best privacy preservation. DR=dECE speaker rank, ER=EER speaker rank. Red: Speakers with EER below 30% across all systems.

Lastly, Table ?? reports **max_abs_LL**R values across all conditions. Values range from 1.415 to 2.389. The lowest one is observed for NAC on **test-m** (1.415, category B) and the highest for ASRBN on **test-f** (2.389, category C). NAC most frequently falls into category B, STTTS predominantly into category C, while ASRBN alternates between B and C across trial sets. The results indicate that strong global performance does not preclude worst-case individual evidence leakage.

The corresponding ZEBRA ECE curves for all three systems and trial partitions are visualized in Appendix ??.

Trial set	System	max_abs_LLRL	Category
dev-f	B3	2.219	C
	B4	1.617	B
	B5	1.793	B
dev-m	B3	2.039	C
	B4	2.340	C
	B5	1.696	B
test-f	B3	1.979	B
	B4	1.787	B
	B5	2.389	C
test-m	B3	2.169	C
	B4	1.415	B
	B5	2.022	C

Table 5: Maximum absolute LLR values and qualitative leakage category across trial sets and anonymization systems.

5.2 Results for Linkability & Singling Out

The Linkability results in Table ?? and Figure ?? follow the same trend reported by ?. That is, Linkability decreases as N' (number of speakers in

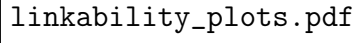
The figure area is mostly blank, with the text 'linkability_plots.pdf' located on the left side.

Figure 1: Linkability as a function of the number of enrollment speakers N' for different conversation lengths L , for all three systems STTTS, NAC and ASRBN.

the enrollment pool from whom the attacker has to choose from) increases, while the metric also scores higher with longer conversation lengths L (meaning the number of utterances used to average the anonymized speaker’s embedding profile).

L	N'	Original	ASRBN	STTTS	NAC
1	5	0.9690 ± 0.0150	0.6198 ± 0.0762	0.6698 ± 0.0420	0.7069 ± 0.0475
	10	0.9526 ± 0.0132	0.4776 ± 0.0572	0.5690 ± 0.0386	0.5819 ± 0.0566
	15	0.9422 ± 0.0166	0.4155 ± 0.0418	0.4897 ± 0.0691	0.5302 ± 0.0396
	20	0.9431 ± 0.0190	0.3741 ± 0.0503	0.4388 ± 0.0563	0.4733 ± 0.0521
	25	0.9319 ± 0.0221	0.3405 ± 0.0452	0.4181 ± 0.0602	0.4190 ± 0.0619
	30	0.9284 ± 0.0251	0.3267 ± 0.0501	0.3897 ± 0.0546	0.4448 ± 0.0682
	40	0.9241 ± 0.0184	0.2784 ± 0.0476	0.3690 ± 0.0618	0.3638 ± 0.0478
	50	0.9103 ± 0.0208	0.2595 ± 0.0588	0.3190 ± 0.0686	0.3457 ± 0.0464
3	5	0.9767 ± 0.0099	0.7388 ± 0.0515	0.7810 ± 0.0405	0.8500 ± 0.0408
	10	0.9672 ± 0.0132	0.6466 ± 0.0699	0.6784 ± 0.0540	0.7440 ± 0.0580
	15	0.9560 ± 0.0207	0.5793 ± 0.0417	0.6440 ± 0.0420	0.7147 ± 0.0441
	20	0.9474 ± 0.0228	0.5371 ± 0.0575	0.5853 ± 0.0474	0.6810 ± 0.0458
	25	0.9388 ± 0.0200	0.5224 ± 0.0554	0.5767 ± 0.0376	0.6474 ± 0.0464
	30	0.9362 ± 0.0238	0.5034 ± 0.0403	0.5526 ± 0.0380	0.6509 ± 0.0393
	40	0.9302 ± 0.0221	0.4690 ± 0.0596	0.5224 ± 0.0482	0.5784 ± 0.0524
	50	0.9284 ± 0.0175	0.4310 ± 0.0488	0.4957 ± 0.0534	0.5534 ± 0.0493
10	5	0.9741 ± 0.0128	0.8388 ± 0.0350	0.8586 ± 0.0265	0.9026 ± 0.0337
	10	0.9612 ± 0.0143	0.7483 ± 0.0461	0.7560 ± 0.0290	0.8681 ± 0.0310
	15	0.9586 ± 0.0127	0.7207 ± 0.0339	0.7086 ± 0.0373	0.8190 ± 0.0387
	20	0.9526 ± 0.0171	0.6810 ± 0.0458	0.6828 ± 0.0246	0.7905 ± 0.0394
	25	0.9491 ± 0.0176	0.6698 ± 0.0359	0.6586 ± 0.0352	0.7733 ± 0.0390
	30	0.9422 ± 0.0166	0.6414 ± 0.0297	0.6284 ± 0.0347	0.7733 ± 0.0315
	40	0.9379 ± 0.0138	0.6009 ± 0.0457	0.6250 ± 0.0288	0.7474 ± 0.0460
	50	0.9293 ± 0.0163	0.5776 ± 0.0384	0.6009 ± 0.0274	0.7147 ± 0.0364
30	5	0.9670 ± 0.0134	0.8193 ± 0.0410	0.8159 ± 0.0366	0.9159 ± 0.0314
	10	0.9614 ± 0.0178	0.7330 ± 0.0448	0.7318 ± 0.0334	0.8670 ± 0.0375
	15	0.9545 ± 0.0190	0.6773 ± 0.0480	0.6898 ± 0.0361	0.8284 ± 0.0233
	20	0.9398 ± 0.0230	0.6614 ± 0.0336	0.6705 ± 0.0384	0.7989 ± 0.0307
	25	0.9227 ± 0.0182	0.6375 ± 0.0309	0.6466 ± 0.0264	0.7852 ± 0.0325
	30	0.9239 ± 0.0194	0.6216 ± 0.0281	0.6239 ± 0.0254	0.7727 ± 0.0321
	40	0.9216 ± 0.0183	0.5875 ± 0.0290	0.6068 ± 0.0250	0.7455 ± 0.0310
	50	0.9068 ± 0.0142	0.5750 ± 0.0217	0.5898 ± 0.0183	0.7318 ± 0.0223

Table 6: Linkability results for conversation lengths L and enrollment candidate sizes N' . Values are mean $\pm$ std across random runs. Lower values indicate lower linkability risk.

As seen from Table ?? and further illustrated in Figure ??, the original condition (meaning the original matching of speakers' utterances with the enrollment profiles) remains highly linkable throughout the different experimental conditions. For example, for $L = 1$, Linkability decreases only from 0.9690 to 0.9103 as N' grows from 5 to 50. Compared to the original setting, anonymization systems substantially reduce Linkability. At $L = 1$ and $N' = 50$, scores drop to 0.2595 for ASRBN, 0.3190 for STTTS, and 0.3457 for NAC.

Moreover, the effect of conversation length is clearly visible both in the table and the figure diagram. For ASRBN at $N' = 50$, Linkability rises from 0.2595 at $L = 1$ to 0.4310 at $L = 3$, then notes 0.5776 at $L = 10$, and 0.5750 at $L = 30$. This trend confirms that aggregating more utterances increases the stability of speaker-specific information. In the figure, the same pattern is observed, with the anonymized systems showing substantially higher scores for $L = 10$ than for $L = 1$.

ASRBN consistently produces the lowest Linkability scores, while NAC shows the highest residual risk, especially at longer conversation lengths. At $L = 30$ and $N' = 50$, Linkability scores are 0.5750 for ASRBN, 0.5898 for STTTS, and 0.7318 for NAC. An interesting notice is that there is a reversal from the EER and ZEBRA rankings where NAC appeared as the second best system.

Singling Out results in Table ?? and Figure ?? follow the same pattern. The original condition shows consistently high isolation risk across all settings, remaining between 0.8716 and 0.9110 for $L = 1$ and above 0.92 for $L = 3$ and $L = 10$.



singling_out_plots.pdf

Figure 2: Singling Out score as a function of speaker population size N for different conversation lengths L . The dotted line indicates the random predicate $\exp(-1) \approx 0.3679$.

L	N	Original	ASRBN	STTTS	NAC
1	20	0.9110 ± 0.0159	0.3790 ± 0.0337	0.4020 ± 0.0286	0.3880 ± 0.0196
	30	0.8820 ± 0.0232	0.3760 ± 0.0273	0.4020 ± 0.0251	0.4000 ± 0.0165
	40	0.8895 ± 0.0186	0.3910 ± 0.0288	0.3945 ± 0.0138	0.3860 ± 0.0222
	50	0.8716 ± 0.0124	0.3676 ± 0.0105	0.4000 ± 0.0083	0.3928 ± 0.0179
3	20	0.9413 ± 0.0378	0.4642 ± 0.0333	0.5368 ± 0.0404	0.5052 ± 0.0863
	30	0.9370 ± 0.0231	0.4264 ± 0.0278	0.5009 ± 0.0438	0.4693 ± 0.0369
	40	0.9293 ± 0.0319	0.3940 ± 0.0535	0.5120 ± 0.0258	0.4320 ± 0.0143
	50	0.9292 ± 0.0133	0.3978 ± 0.0242	0.4798 ± 0.0167	0.4320 ± 0.0333
10	20	0.9533 ± 0.0194	0.5533 ± 0.0287	0.6033 ± 0.0662	0.7267 ± 0.0573
	30	0.9756 ± 0.0178	0.4867 ± 0.0285	0.5911 ± 0.0447	0.6289 ± 0.0736
	40	0.9233 ± 0.0153	0.4717 ± 0.0427	0.5783 ± 0.0239	0.6083 ± 0.0444

Table 7: Singling Out results for conversation lengths L and speaker population sizes N . Values are mean $\pm$ std across random runs. Random-predicate baseline: $\exp(-1) \approx 0.3679$.

All anonymization systems substantially seem to reduce Singling Out compared to the original utterances. More specifically, at $L = 1$ and $N = 50$, scores drop to 0.3676 for ASRBN, 0.4000 for STTTS, and 0.3928 for NAC, approaching the random baseline of $\exp(-1) \approx 0.3679$. As L increases, scores rise above the random baseline for all systems: for ASRBN at $N = 20$, Singling Out increases from 0.3790 at $L = 1$ to 0.4642 at $L = 3$ and 0.5533 at $L = 10$. ASRBN again provides the strongest protection, while NAC shows the highest residual risk at longer conversation lengths. At $L = 10$ and $N = 20$, scores are 0.5533 for ASRBN, 0.6033 for STTTS, and 0.7267 for NAC, further supporting the Linkability finding that NAC preserves more residual speaker information than EER and ZEBRA suggest.

5.3 Phonetic, Acoustic, and Linguistic Factors in Speaker Re-identification

This section reports the results of the phonetic and acoustic feature analysis which are presented in four steps:

1. Target minus non-target feature space separation is used to test whether each feature space assigns higher similarity to same-speaker trials than to different speaker trials (Table ??).
2. Trial-level correlations with ASV scores (Table ??).
3. Predictive models are used to test how much ASV-score variation can be explained by linguistic and phonetic/acoustic features (Table ??).
4. Speaker-level correlations with individual EER are reported to explore which specific features may be associated with speaker vulnerability (Table ??).

Initially, in Table ?? the clearest evidence that the phonetic feature spaces preserve speaker-discriminative structure is provided. There, larger positive values indicate that the feature space assigns higher similarity to the same-speaker trials than to different-speaker trials. Across all trial sets and systems, rhythm, phone duration and trajectory features show consistently positive target minus non-target separation. Overall, phone duration and trajectory features show the strongest separations, for example in the `test-m` condition phone duration separation reaches 0.436 for STTTS, 0.442 for NAC and 0.406 for ASRBN, while trajectory separation reaches 0.421, 0.455 and 0.415, respectively. Rhythm also shows consistent separation but slightly lower than the aforementioned collective features.

In contrast, openSMILE (which includes the combination of the 89 extracted features in the library) and ASV-embedding similarity show much weaker separation. openSMILE values remain close to zero, with the largest value being 0.075 for `test-f` STTTS, while ASV-embedding similarity is also close to zero or slightly negative in most cases. This suggests that rhythm, phone duration and trajectory,

retain more same-speaker structure than the broader acoustic or embedding-based similarity measures. This means that after anonymization, same-speaker trial pairs remain more similar than different-speaker trial pairs in the phonetic feature spaces of rhythm, phone durations and trajectory features. However, this does not necessarily mean that these features alone are sufficient for re-identification, but it rather shows that they contain speaker-discriminative information that may provide the attacker with cues.

Trial set	System	Rhythm	Phone dur.	Trajectory	openSMILE	ASV emb.
dev-f	STTTS	0.272	0.381	0.345	0.028	-0.003
	NAC	0.288	0.359	0.400	0.004	0.007
	ASRBN	0.223	0.381	0.406	0.022	0.002
dev-m	STTTS	0.316	0.380	0.373	0.005	0.016
	NAC	0.342	0.362	0.359	0.044	0.007
	ASRBN	0.275	0.369	0.381	0.028	-0.002
test-f	STTTS	0.326	0.348	0.309	0.075	-0.006
	NAC	0.334	0.343	0.360	0.051	-0.000
	ASRBN	0.276	0.347	0.391	0.050	-0.005
test-m	STTTS	0.356	0.436	0.421	0.049	0.021
	NAC	0.366	0.442	0.455	0.060	-0.009
	ASRBN	0.316	0.406	0.415	0.031	-0.002

Table 8: Target minus non-target feature-space cosine similarity separation across systems and trial sets. ASV emb. denotes ASV embedding similarity.

In Table ?? the Spearman correlations between feature-space similarity and ASV scores are reported. The correlations are generally positive but modest. This means that higher phonetic similarity is often associated with higher ASV scores, but the relationship is not strong enough to fully explain the ASV system’s behaviour. Phone duration similarity shows the most consistent positive association with ASV scores, especially for ASRBN, where it reaches 0.221 on **dev-f**, 0.224 on **dev-m** and 0.218 on **test-m**. Rhythm also shows moderate positive correlations in several conditions, such as 0.185 for **dev-m** ASRBN and 0.170 for **test-f** NAC.

Trial set	System	Rhythm	Phone dur.	Trajectory	openSMILE	ASV emb.
dev-f	STTTS	0.090	0.128	0.065	0.009	0.102
	ASRBN	0.155	0.221	0.062	-0.002	0.051
	NAC	0.149	0.203	0.047	-0.043	-0.023
dev-m	STTTS	0.165	0.147	0.079	-0.008	0.203
	ASRBN	0.185	0.224	0.074	0.046	0.066
	NAC	0.167	0.175	0.051	0.003	0.017
test-f	STTTS	0.111	0.102	0.059	0.046	0.071
	ASRBN	0.146	0.148	0.044	-0.035	-0.126
	NAC	0.170	0.142	0.053	-0.020	-0.049
test-m	STTTS	0.037	0.067	0.155	-0.035	0.202
	ASRBN	0.173	0.218	0.036	0.000	0.021
	NAC	0.163	0.147	0.003	-0.024	-0.090

Table 9: Spearman correlations between feature-space similarity and ASV score across all trials. ASV emb. denotes ASV embedding similarity.

Trajectory correlations with ASV scores are weaker and less consistent, except for **test-m** STTTS, where trajectory reaches 0.155. openSMILE correlations are close to zero or negative in most conditions, suggesting that the openSMILE similarity measure used here does not strongly track ASV scores. ASV-embedding similarity is also inconsistent, with low positive values for some STTTS conditions but negative values for NAC in **dev-f**, **test-f** and **test-m**.

Overall, this table suggests that phonetic similarity is related to ASV scoring, particularly through phone duration and rhythm, but only partially. Phone duration preservation may be contributing to the way target and non-target trials are matched by the ASV system. However, the correlations remain modest, meaning that phone duration similarity is not sufficient on its own to explain ASV score behaviour. So the stronger evidence for residual phonetic speaker structure comes from the target minus non-target separation in Table ??, while Table ?? shows that this structure only weakly transfers to ASV similarity score prediction.

Next, the predictive contribution of linguistic, phonetic/acoustic and their in-

interaction feature groups is indicated in Table ?? . The R^2 values show that phonetic/acoustic features alone explain only a limited proportion of the ASV-score variance. Then, the phonetic/acoustic-only R^2 values, ranging from 0.024 to 0.077, indicate that these features are not sufficient on their own to model the ASV score. Text-only features generally explain more variance than phonetic/acoustic-only features for NAC and ASRBN, especially in the female trial sets. For example, in **dev-f** ASRBN, the text-only model reaches $R^2 = 0.123$, while the phonetic/acoustic-only model reaches $R^2 = 0.061$.

The combined models consistently outperform the individual feature groups. This indicates that text-based and phonetic/acoustic information contribute complementary information to ASV-score prediction. For example, in **test-m** ASRBN, the combined model reaches $R^2 = 0.176$, compared with 0.119 for text-only and 0.057 for phonetic/acoustic-only. A similar pattern is observed for NAC, where the combined model reaches 0.160 on **test-m**, compared with 0.121 for text-only and 0.041 for phonetic/acoustic-only. However, adding interaction terms produces only very small improvements over the combined model. This suggests that the interaction between text similarity and phonetic/acoustic similarity does not improve the ASV-score prediction beyond the additive combination of both feature groups.

Last but not least, the best AUC column shows that the same feature groups are more successful at distinguishing target from non-target trials than at predicting the exact ASV score. All AUC values are above 0.80, with the highest value observed for **test-m** STTTS (0.868).

Therefore, the feature groups contain enough information to support target/non-target separation within this trial-level evaluation setup, even though they explain only a limited amount of the ASV-score variance. As a result, this table supports the interpretation that phonetic/acoustic features are not sufficient to fully predict ASV scores, but they contribute useful information when combined with text features and support target/non-target classification within the evaluated trial pairs.

Trial set	System	Text only	Phon./Acous. only	Combined	+ Int.	Best AUC
dev-f	STTTS	0.031	0.025	0.051	0.052	0.819
	NAC	0.100	0.053	0.137	0.138	0.826
	ASRBN	0.123	0.061	0.163	0.165	0.815
dev-m	STTTS	0.041	0.069	0.108	0.108	0.838
	NAC	0.103	0.049	0.155	0.155	0.818
	ASRBN	0.068	0.077	0.145	0.147	0.813
test-f	STTTS	0.016	0.024	0.041	0.044	0.801
	NAC	0.105	0.043	0.142	0.144	0.824
	ASRBN	0.115	0.051	0.156	0.157	0.813
test-m	STTTS	0.016	0.069	0.087	0.088	0.868
	NAC	0.121	0.041	0.160	0.163	0.841
	ASRBN	0.119	0.057	0.176	0.176	0.825

Table 10: Five-fold trial-level cross-validated comparison of linguistic and phonetic/acoustic feature groups across anonymization systems and trial sets. The Text only, Phon./Acous. only, Combined, and + Int. columns report R^2 for **ASV score prediction**. Combined denotes the model using both text and phonetic/acoustic features. + Int. denotes the model with text, phonetic/acoustic features, and interaction terms. Best AUC reports the highest ROC-AUC obtained for target/non-target classification.

Table ?? focuses on phonetic preservation features that are negatively correlated with individual speaker EER. Spearman correlation is used because the relationship between phonetic preservation features and EER is not assumed to be linear. The purpose of the table is to identify phonetic properties whose preservation after anonymization may be associated with residual re-identification. However, given the small number of speakers per condition ($n = 13\text{--}16$), the absence of inferential significance testing, and the lack of correction for multiple comparisons, the reported coefficients should be interpreted only as descriptive and exploratory point estimates, rather than as statistically confirmed associations or reliable differences across systems.

Trial set	System	Group	Feature	n	ρ
dev-f	STTTS	Rhythm	Vocalic nPVI preservation	15	-0.164
dev-f	NAC	Phone d.	Phone-duration pattern preservation	15	-0.557
	NAC	Traj.	F3 preservation	15	-0.471
	NAC	Phone d.	Phone-duration pattern preservation	15	-0.243
	NAC	Rhythm	Vocalic rPVI preservation	15	-0.196
	NAC	Rhythm	Mean consonant duration preservation	15	-0.161
dev-f	ASRBN	Rhythm	Vocalic ΔV preservation	15	-0.357
	ASRBN	Traj.	F1 preservation	15	-0.314
	ASRBN	Traj.	F3 preservation	15	-0.296
	ASRBN	Traj.	F0 preservation	15	-0.154
dev-m	STTTS	Rhythm	Percentage of consonantal material preservation	14	-0.510
	STTTS	Traj.	F2 preservation	14	-0.317
dev-m	NAC	Traj.	F2 preservation	14	-0.116
dev-m	ASRBN	Rhythm	Vocalic ΔV preservation	14	-0.371
	ASRBN	Rhythm	Consonantal ΔC preservation	14	-0.270
	ASRBN	Rhythm	Percentage of consonantal material preservation	14	-0.116
	ASRBN	Rhythm	Percentage of vocalic material preservation	14	-0.116
test-f	STTTS	Rhythm	Vocalic nPVI preservation	16	-0.512
	STTTS	Traj.	F2 preservation	16	-0.253
test-f	NAC	Traj.	F0 preservation	16	-0.227
	NAC	Rhythm	Consonantal nPVI preservation	16	-0.139
test-f	ASRBN	Rhythm	Consonantal nPVI preservation	16	-0.468
	ASRBN	Rhythm	Percentage of vocalic material preservation	16	-0.288
	ASRBN	Rhythm	Percentage of consonantal material preservation	16	-0.288
test-m	STTTS	Traj.	F1 trajectory preservation	13	-0.747
	STTTS	Rhythm	Consonantal rPVI preservation	13	-0.401
	STTTS	Rhythm	Vocalic nPVI preservation	13	-0.286
	STTTS	Rhythm	Vocalic Varco preservation	13	-0.247
	STTTS	Rhythm	Consonantal nPVI preservation	13	-0.247
test-m	NAC	Phone d.	Phone-duration pattern preservation	13	-0.527
	NAC	Rhythm	Vocalic rPVI preservation	13	-0.473
	NAC	Traj.	F0 preservation	13	-0.198
	NAC	Rhythm	Vocalic nPVI preservation	13	-0.159
	NAC	Rhythm	Percentage of consonantal material preservation	13	-0.148
test-m	ASRBN	Rhythm	Consonantal rPVI preservation	13	-0.269
	ASRBN	Traj.	F2 preservation	13	-0.264
	ASRBN	Traj.	F3 preservation	13	-0.236
	ASRBN	Rhythm	Vocalic nPVI preservation	13	-0.187

Table 11: Phonetic preservation features negatively correlated (Spearman) with speaker-level EER, by system and trial set. Negative correlations indicate that stronger preservation is associated with lower EER.

The reported features are preservation measures, for example, a rhythm preservation feature indicates that a given rhythm property remains similar between the original and anonymized speech. Similarly, a trajectory preservation feature indicates that the temporal shape of a pitch or formant contour remains similar after anonymization. Only the negative Spearman correlations are reported because they indicate cases where higher preservation of a phonetic feature is associated with lower speaker EER. In simple terms, the more strongly this feature is preserved for a speaker after anonymization, the easier that speaker may be for the attacker to re-identify.

Several types of phonetic features appear in the table. Rhythm features describe the temporal organization of speech. *Vocalic* and *consonantal percentages* measure how much of an utterance is occupied by vowel-like and consonant-like intervals. Mean vowel and consonant durations describe the average duration of these intervals. ΔV and ΔC refer to the standard deviation of vocalic and consonantal interval durations, respectively. *Varco-V* and *Varco-C* are normalized variability measures, computed by normalizing ΔV and ΔC by the corresponding mean vowel or consonant duration. Finally, *rPVI* (raw) and *nPVI* (normalized) measure pairwise variability between successive vocalic or consonantal intervals. Phone-duration features describe the duration pattern of individual phones, and phone-duration pattern preservation means that the relative timing of phones in the anonymized speech remains similar to the original speech. Lastly, trajectory features describe the temporal evolution of acoustic contours. F0 refers to the pitch contour, while F1, F2 and F3 refer to the first three formant trajectories. These formants are related to the resonant structure of speech and are influenced by articulation and vocal tract characteristics.

The strongest negative association in the table appears for **test-m** STTTS, where F1 trajectory preservation has a correlation with EER of $\rho = -0.747$. This indicates that, in this condition, speakers whose F1 trajectory is more preserved after anonymization tend to have substantially lower EER. Therefore, F1 trajectory preservation may act as a residual cue for the attacker in this subset. Another strong negative association is observed for **dev-f** and **test-m** NAC, where phone-duration pattern preservation has $\rho = -0.557$ and $\rho = -0.527$, respectively. This suggests that preserving the original phone-duration pattern may make some speakers more vulnerable under NAC.

Rhythm preservation features also appear very frequently in the table. For example, **test-f** STTTS shows a negative correlation for vocalic nPVI preservation ($\rho = -0.512$), while in ASRBN it shows negative correlations for consonantal nPVI preservation ($\rho = -0.468$) and percentage-based rhythm preservation. Rhythm-related features are the most frequent category overall, appearing across almost all systems and trial sets. This suggests that timing structure, especially the relative organization of vocalic and consonantal intervals, may carry residual speaker-specific information after anonymization.

Trajectory features are the second most frequent category. They are especially important in **test-m** STTTS, where F1 trajectory preservation shows the strongest negative correlation in the table, and in **dev-f** ASRBN and **test-m** ASRBN, where formant trajectory preservation also appears. This indicates that the preservation of pitch or formant contour shapes may also contribute to re-identification vulnerability in some conditions. Phone-duration preservation appears less frequently than rhythm and trajectory features, but for NAC phone-duration pattern preservation is strongly negatively correlated with EER in both **dev-f** and **test-m**.

The speaker-level heatmaps in Appendix ?? provide a visual complement to Table ??, showing that some vulnerable speakers (top rows, lowest EER) exhibit above-average preservation in phone-duration and trajectory feature spaces. This trend, nevertheless, is not constant across all speakers and systems. Speaker **6241** in B3 **dev-m**, for instance, retains high preservation across nearly all feature spaces despite a mid-range EER, while speaker **84** in NAC **dev-f** is among the best-protected

speakers overall ($\text{EER} = 61.1\%$) yet shows strongly above-average preservation in phone-duration and rhythm, and then speaker **6241** in ASRBN `dev-m` shows a similar mismatch in phone-duration and rhythm. On the other hand there are speakers with quite low EER scores like **7021** in ASRBN `test-m` or **8297** in STTTS, which do not show consistently high preservation values across the examined feature spaces. These cases indicate that the link between phonetic preservation and vulnerability is a general tendency rather than a strict rule. However, once again, these results should be interpreted carefully, because the number of available speakers per condition is small. Therefore, the correlations are exploratory and neither the table nor the heatmaps prove that these features cause re-identification but they only indicate that phonetic preservation and speaker vulnerability tend to co-occur more often than not.

6 Discussion

This thesis is formed on the basis of four questions. First, it asks in which level the three selected VPC 2024 baseline systems, STTTS, NAC, and ASRBN, preserve privacy across individual speakers based on the official adopted by VPC metric, EER, and whether their differences become visible (**RQ1**, Section ??). Second, it examines whether alternative privacy metrics proposed in literature, i.e. ZEBRA, Linkability & Singling Out, provide a more detailed view of anonymization performance than EER alone (**RQ2**, Sections ?? and ??). Third, it investigates whether speaker-level vulnerability after anonymization is associated with factors such as gender, phoneme duration, pitch, rhythm characteristics, formant trajectories and linguistic content (**RQ3**, Section ??). Finally, it asks what these findings reveal about the remaining weaknesses of current anonymization systems and what should be taken into account in order to make voice anonymization systems and their evaluation more robust in the future (**RQ4**, partially in Section ?? and mainly in Section ??).

The results presented in the previous section address these questions. Taken together, they show that privacy performance after voice anonymization cannot be fully understood through a single global score and even though the systems differ in

their overall behaviour (ASRBN often achieving the strongest global privacy performance and STTTS the weakest) the important finding is that some speakers remain comparatively easy to re-identify across systems with different architectures, while others appear to be protected more effectively, confirming the literature findings (??) and further supporting the statement that global EER can hide individual-level differences. The alternative metrics refine rather than simply reproduce this picture: ZEBRA shows that speakers with similar EER values may still differ in the amount of residual identity evidence available to the attacker based on **dECE**, while Linkability & Singling Out shift the interpretation to the possible linkage or isolation of anonymized speech samples. Lastly, the feature-level analyses add interpretation to the metrics’ results by showing that residual recognizability seems to be present and related to the preservation of specific phonetic/acoustic and linguistic properties, though this relationship does not seem to be consistent across systems and trial sets, thus strong claims should be avoided.

6.1 Baseline Performance and EER as an Incomplete Privacy Metric

The results in Section ?? show that global EER, although convenient, can give an incomplete picture of anonymization effectiveness. Across all trial sets and systems, global EER values fall within or close to the VPC privacy thresholds. ASRBN achieves the strongest global results, with EER values of 34.231% on **dev-f** and 32.770% on **test-m**, while STTTS performs the weakest, with 28.265% and 26.278% on the same sets. At the population level, these results suggest meaningful privacy protection.

However, the speaker-level EER analysis reveals persistent individual vulnerabilities. On **dev-f**, speaker **1673** obtains low EER values across all systems, with 10.078%, 18.651%, and 14.515% under STTTS, NAC, and ASRBN respectively. Speaker **2412** shows a similar pattern, remaining below 30% EER for all systems. On **dev-m**, speakers **652**, **2803**, and **5536** are also consistently vulnerable, with speaker **2803** reaching EER values as low as 9.174%, 18.946%, and 17.102%. Sim-

ilar cases appear in the test sets, including speakers **3570** and **1221** on **test-f**, and speakers **1320**, **4077**, **7021**, and **8224** on **test-m**. The fact that these speakers remain vulnerable across fundamentally different architectures suggests that the failures are not only system-specific, but may also reflect persistent speaker-dependent properties. This supports previous findings by ?, who identified speakers that remain difficult to anonymize across systems, and by ?, who showed that per-speaker EER can be very low even when global EER appears acceptable.

From a privacy-law perspective, such individual failures are important. Under Recital 26 of the GDPR, information is anonymous only when the data subject is no longer identifiable, not merely when most speakers are protected (?). Global EER cannot capture these selective failures because it summarizes performance across speakers. This motivates the use of complementary metrics that also expose individual-level privacy leakage.

6.2 What ZEBRA Adds: Its Divergence from EER

The ZEBRA framework with its complementary measures of privacy disclosure operates on the evidence produced by the ASV system rather than on a single operating point threshold (as described in Section ??). With the **dECE** quantifying how much identity evidence in bits the anonymized speech provides to the adversary, and the **max_abs_LL** capturing the strongest individual piece of evidence observed among all trial scores this framework exposes dimensions of privacy that the EER cannot capture.

At the global level, the ordering of systems under **dECE** is consistent with their ordering under EER. ASRBN achieves the lowest global **dECE** across three of the four trial sets, corresponding to its generally highest global EER, while STTTS consistently produces the highest global **dECE**, corresponding to its lowest global EER. This might suggest that ZEBRA and EER are acting similarly, but the speaker-level analysis reveals that this consistency does not extend to the individual level, and that the two metrics capture meaningfully different aspects of privacy disclosure.

Several speakers show mismatches between speaker-level EER and dECE. For example, on `dev-f` under NAC, speaker **6313** has a low EER of 19.949%, suggesting weak protection, but a relatively low dECE of 0.087, indicating limited overall evidence leakage. In contrast, speaker **3536** has an EER of 30.278%, which appears more acceptable, but a higher dECE of 0.201. Speaker **2412** shows a similar pattern, with an EER close to 30% but a dECE of 0.243. Similar cases appear on `test-f` under NAC, where speaker **2961** has an EER of 30.407% but a dECE of 0.274, and on `test-m` under ASRBN, where speaker **61** has an EER of 30.244% but a dECE of 0.156. These cases are only some examples that reflect the different nature of the two metrics. EER measures only a point but it does not describe the full score distribution or measure the identity leakage, unlike dECE. Moreover, the inherent inclusion of `max_abs_LL`R in the framework highlights the worst-case leakage by capturing high-risk individual trials that are masked by EER.

6.3 Linkability & Singling Out: A Different Threat Assessment Metric

The Linkability & Singling Out results in Section ?? provide a different view of privacy risk, closer to the legal notion of anonymization. Unlike EER and ZEBRA, which are based on ASV similarity scores, Linkability evaluates whether anonymized trial embeddings can still be associated with the correct enrollment speaker within a candidate pool and Singling Out evaluates whether an attacker can isolate one individual from a set of anonymized representations, therefore capturing residual speaker information that may not appear as strong ASV performance, but still represent a privacy risk under the GDPR framework.

Here, the effect of conversation length seems of high importance across all systems. As L increases, Linkability also increases, showing that more speech material produces more stable speaker representations. For example, for NAC at $N' = 50$, Linkability rises from 0.3457 at $L = 1$ to 0.7318 at $L = 30$. The Singling Out results follow the same tendency with systems close to the random predicate baseline at $L = 1$, but clearly exceed it at longer conversation lengths. This supports the idea

that repeated speech collection creates a realistic privacy risk, because an attacker can aggregate anonymized utterances over time.

The main finding is that system ranking changes when these metrics are used. While NAC ranks second under EER and ZEBRA, it shows the highest residual risk under Linkability and Singling Out across most conditions. For example, at $L = 30$ and $N' = 50$, NAC reaches a Linkability score of 0.7318, compared with 0.5750 for ASRBN and 0.5898 for STTTS. Similarly, at $L = 10$ and $N = 20$, NAC reaches a Singling Out score of 0.7267, compared with 0.5533 for ASRBN and 0.6033 for STTTS. This reversal shows that a system can perform well under verification-based metrics while still preserving information that allows utterances from the same speaker to be linked over time.

This behaviour could be connected to NAC’s token-based architecture. In this approach, semantic tokens are extracted from the source utterance, while acoustic prompts from pseudospeakers condition the generation of acoustic tokens for the anonymized waveform, with the aim to suppress speaker information (?). However, the higher Linkability scores observed here suggest that some stable speaker-specific regularities may remain in the generated token sequences. When multiple utterances are aggregated, these residual regularities become more stable, which could explain why NAC is more vulnerable under Linkability, especially for longer conversation lengths.

ASRBN performs best under both Linkability & Singling Out, which is consistent with the role of vector quantization in its architecture. As ASR bottleneck features are not inherently speaker-independent, they can still leak speaker identity (?), so by constraining the ASR bottleneck representation to a limited set of discrete prototypes, vector quantization reduces the amount of speaker-specific variation preserved in the linguistic features. This may explain why ASRBN remains less linkable than NAC and STTTS when several utterances are aggregated.

Overall, these results show that privacy evaluation should not rely only on EER. Linkability & Singling Out capture the accumulation of speaker evidence across repeated utterances, which is highly relevant for real-world deployments. Their in-

clusion in future evaluation pipelines, as suggested by ?, would make system comparison more aligned with both GDPR oriented privacy requirements and practical attack scenarios.

6.4 The Role of Gender and Phonetic/Linguistic Features as Residual Identity Cues

The phonetic, acoustic, and linguistic analysis in Section ?? shows that anonymization does not fully remove speaker-discriminative structure from the speech signal. As shown in Table ??, rhythm, phone-duration, and pitch/formant trajectory features consistently produce positive target minus non-target separation across all systems and trial sets, with the strongest separations observed for phone-duration and trajectory features.

The trial-level correlations in Table ?? further support this interpretation, with phone-duration similarity showing the most consistent positive association with ASV scores, especially for ASRBN, where it reaches 0.221 on `dev-f`, 0.224 on `dev-m`, and 0.218 on `test-m`. Rhythm also correlates positively in several conditions, for example 0.185 for ASRBN on `dev-m` and 0.173 on `test-m`. These correlations are modest, so phonetic preservation should not be treated as the only factor driving ASV scores. Rather, they indicate that timing and phonetic structure partially overlap with the information exploited by the attacker.

To continue at the speaker level, Table ?? shows that stronger preservation of specific phonetic features is often associated with lower speaker-level EER, meaning that speakers whose anonymized speech preserves these features more strongly tend to be more vulnerable. The descriptive patterns differ by system, with different feature categories appearing more prominently across conditions.

For STTTS, rhythm and trajectory features are prominent. For example, on `test-m`, F_1 trajectory preservation has a strong negative correlation with EER ($\rho = -0.747$), while consonantal rPVI preservation also correlates negatively ($\rho = -0.401$). This is plausible given the architecture of STTTS, which retains origi-

nal phonetic transcriptions and phone durations unmodified, and only randomly perturbs pitch and energy at the phone level ?. It is interesting how in ? unmodified prosody cloning (duration, pitch, and energy) does not by itself compromise privacy, provided the substituted speaker embedding is sufficiently dissimilar from the original. This suggests the possibility that the descriptive associations observed here are related to specific formant and rhythm properties that remain after B3’s perturbation scheme, rather than to prosody preservation in general.

For NAC, phone-duration preservation provides some of the largest negative descriptive correlations with speaker-level EER on **dev-f**. Also, **test-m** has a correlation with EER of ($\rho = -0.527$). This could be possibly interpreted based on the fact that NAC is not explicitly designed to modify phone durations (?). The results therefore suggest that some segmental timing regularities may survive the token-based transformation and become useful when comparing utterances from the same speaker.

For ASRBN, rhythm-related features appear more frequently among the reported negative descriptive correlations. For example, vocalic ΔV preservation correlates with lower EER on **dev-f** ($\rho = -0.357$) and **dev-m** ($\rho = -0.371$), while consonantal nPVI preservation reaches $\rho = -0.468$ on **test-f**. This fits the motivation of ASRBN only in a cautious sense, as ASR bottleneck features may still contain speaker information (?).

The linguistic analysis in Table ?? adds another layer to these interpretations. Text-only features seem to explain more ASV-score variance than phonetic/acoustic features alone for NAC and ASRBN in several trial sets. This trend seems constant with the findings of ?, ? and ? who show that similar linguistic content in enrollment and trial embeddings can be a strong residual identity cue. For example, on **dev-f**, ASRBN reaches $R^2 = 0.123$ with text-only features compared with 0.061 for phonetic/acoustic features, and NAC reaches 0.100 compared with 0.053. For STTTS, the contribution is more balanced, and phonetic/acoustic features are sometimes stronger, for example on **dev-m** and **test-m**, mainly because STTTS retains the prosody of the speaker in the anonymized version. Across all systems, the combined model outperforms either feature group alone. The interaction terms add

only small improvements, suggesting that linguistic and phonetic/acoustic information contribute mostly additively and not in an interactive way to re-identification, not supporting the claim that *what is said* is relevant to *how it is said*.

The gender-partitioned results further show that vulnerability is present in both female and male speaker sets, but it is not distributed identically across them. At the global level, the male partitions generally show lower EER values than the female partitions, especially for STTTS. For example, STTTS decreases from 28.265% on `dev-f` to 23.788% on `dev-m`, and from 28.682% on `test-f` to 26.278% on `test-m`. A similar tendency appears for NAC and ASRBN on the development sets, where both systems obtain higher EER on `dev-f` than on `dev-m`. The speaker-level results also show that several male speakers remain below the 30% EER threshold across all systems, such as speakers **652**, **2803**, and **5536** on `dev-m`, and speakers **1320**, **4077**, **7021**, and **8224** on `test-m`. However, this pattern is not exclusive to male speakers, since female speakers such as **1673**, **2412**, **3570**, and **1221** also remain consistently vulnerable across systems. These results should not be interpreted as showing a simple gender effect, but rather as indicating that gender-partitioned evaluation can reveal differences in how vulnerability is distributed across speaker groups.

6.5 Implications for Future Anonymization Systems and their Evaluation

The results suggest that future anonymization systems should not aim solely on increasing global EER, but should also adopt the introduction of a broader set of metrics. EER still remains useful for comparison with previous VPC results, but the findings of this thesis show that it can hide individual-level vulnerability and metric-dependent changes in system ranking. Linkability & Singling Out and the ZEBRA framework also need to be considered as they incorporate metrics which are based on the legal and forensic frameworks respectively, and therefore correspond to different aspects of privacy protection. Especially in real-life scenarios where an attacker may collect several anonymized utterances from the same person over time, these metrics

provide a more realistic assessment of privacy risk than EER alone.

Linguistic content can also function as a leakage channel in this experimental setup, yet remains a largely untouched topic by the challenge organisers. The current systems anonymize only *how* something is said, not *what* is said. Future VPC occurrences should aim for testing other types of datasets with data that imitate real-life scenarios as ? suggest. Moreover, stronger privacy guarantees may need to consider content-level interventions, or at least explicitly report the probability of this leakage rather than treating it as a solely corpus artefact that the system failed to conceal.

Finally, the results indicate that anonymization should be tested across speaker groups like ? and not only through average scores. Some speakers remain vulnerable across all three systems, which suggests that future work should include larger-scale and targeted analyses of hard to anonymize speakers and use these cases to further guide system design. As far as the scope of this thesis is concerned, B5 or ASRBN appears to be the most robust system overall according to all the tested metrics and their individual-level analyses. This is in line with ? that constraining ASR bottleneck features with vector quantization substantially reduces speaker linkability since a sufficiently small prototype dictionary forces the network to encode a more speaker-invariant linguistic representation. He showed also that F0 transformations alone provide little privacy gain when the main leakage source lies in the linguistic bottleneck rather than in pitch, further also supporting the advocates of linguistic leakage. Given ASRBN’s comparatively strong and consistent performance across all metrics used in this thesis, future work should prioritize refining this quantization-based approach, for instance by extending it to rhythm and duration representations (which emerged as candidate residual factors in the exploratory analysis in Table ??), rather than moving away from it. So, a more robust anonymization pipeline would therefore combine a wider, real-life based and naturalistic evaluation dataset with an ASRBN style bottleneck quantization approach extended to more strongly suppress rhythmic cues, evaluated with methods that reveal both population-level and speaker-level privacy risks. This would be a promising direction for future experimentation.

7 Conclusions and Future Work

This thesis examines alternative metrics for assessing the performance of voice anonymization systems, highlights the shortcomings of global EER, tests different metrics available and more suitable for this task and lastly investigates if the residual speaker vulnerability can be associated with phonetic/acoustic features and their combination with linguistic information. Three VPC 2024 baseline systems, STTTS, NAC, and ASRBN, were evaluated using global and speaker-level EER, ZEBRA, Linkability, and Singling Out, together with the feature-level analysis to provide explainability for the obtained results.

The results showed that global privacy performance can conceal substantial differences between individuals. Although ASRBN generally achieved the strongest privacy protection and STTTS the weakest, several speakers remained vulnerable across all three systems. ZEBRA further showed that speakers with similar EER values may differ in the amount of identity evidence disclosed. Linkability & Singling Out revealed that anonymized speech became easier to associate and isolate when several utterances were aggregated. In particular, NAC showed greater residual risk under these metrics than its EER results suggested.

The feature analysis indicated that rhythm, phone-duration patterns, and pitch and formant trajectories preserve speaker-discriminative structure after anonymization. Linguistic similarity also contributed to ASV-score prediction, especially when combined with phonetic and acoustic information. However, these findings were not fully consistent across systems and partitions and should therefore be interpreted as exploratory, given also the limited number of speakers in the trial sets.

Overall, the findings demonstrate that voice anonymization should not be assessed through global EER alone given that there are evaluation frameworks available that combine population-level and speaker-level measures and which are evidence-based, correspond to real-life scenarios or are legally grounded.

Future work should repeat the application of metrics and the feature analyses using larger speaker populations, different languages, taking also into account the

latest VPC of 2026 (?), and more naturalistic datasets containing spontaneous and conversational speech that reflect real-life application scenarios. This would help determine whether the observed rhythm, duration, trajectory, and linguistic effects generalize beyond LibriSpeech.

A recently proposed metric that would be especially relevant for future evaluation is Similarity Rank Disclosure (SRD) (?). SRD measures privacy disclosure directly from feature representations by ranking the matching speaker among a set of reference speakers. It is independent of classifier thresholds and reports both average and worst-case disclosure in information bits. It can also be applied to different representations, including speaker embeddings, fundamental frequency, and phone-based features which is useful because it provides a multidimensional view of privacy, which in this work I tried to do with the feature-level analyses. Future work could therefore compare its findings with EER, ZEBRA, Linkability & Singling Out, test them after performing anonymization on larger, more naturalistic datasets and see if the results align with the trends noticed in this thesis.

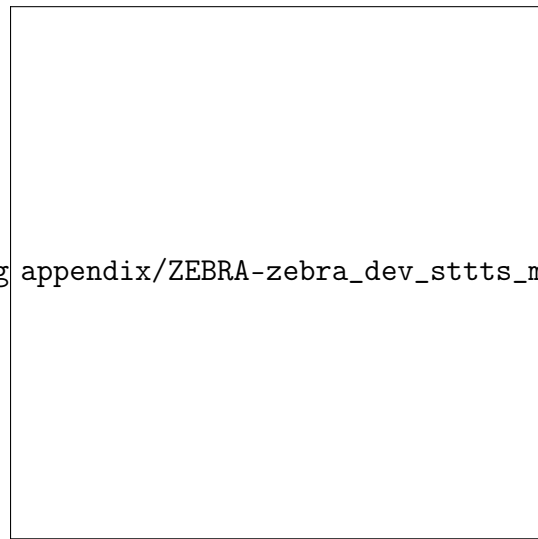
A Appendix

A.1 ZEBRA Curves on Partition Sets

This section presents additional ZEBRA ECE curves across systems and partition sets. The black curve indicates the zero-evidence reference, while the colored curve shows the calibrated ECE profile. The legend summarizes $\mathbf{dECE}$, maximum absolute LLR, and the ZEBRA evidence category.

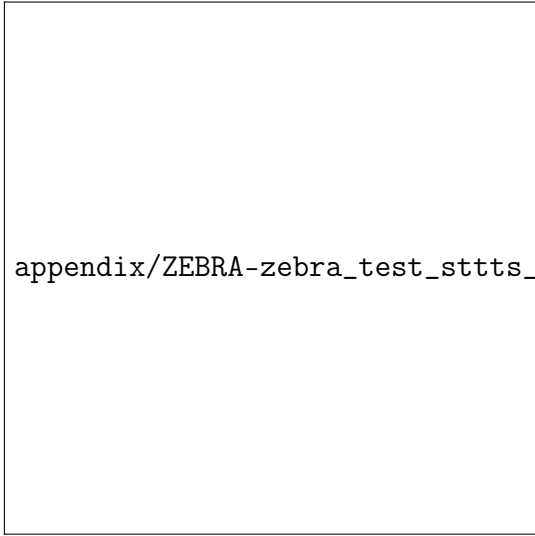


(a) STTTS, dev-f

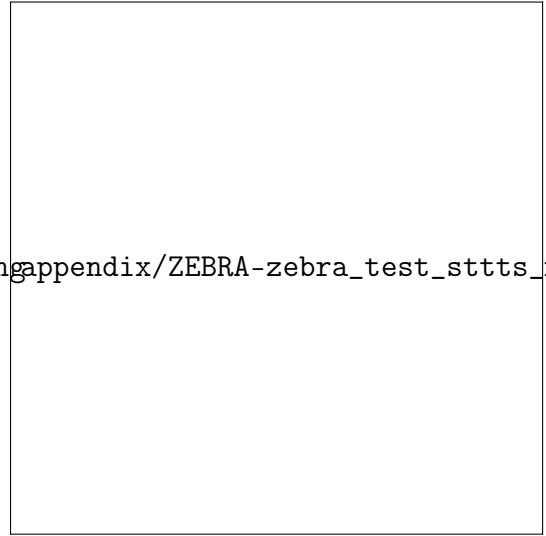


(b) STTTS, dev-m

Figure 3: ZEBRA ECE curves for STTTS on the development partitions.



(a) STTTS, test-f



(b) STTTS, test-m

Figure 4: ZEBRA ECE curves for STTTS on the test partitions.

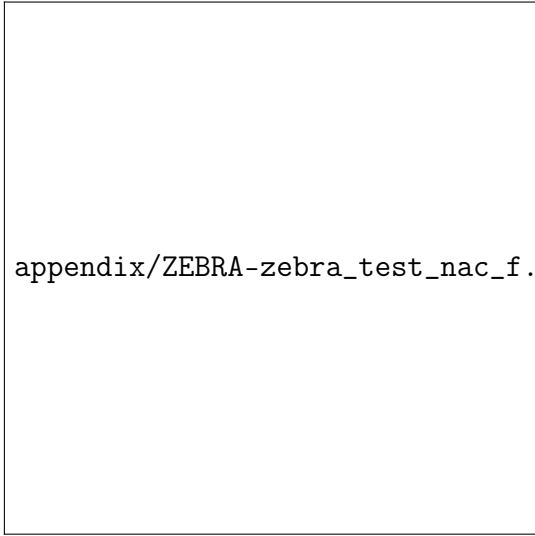


(a) NAC, dev-f

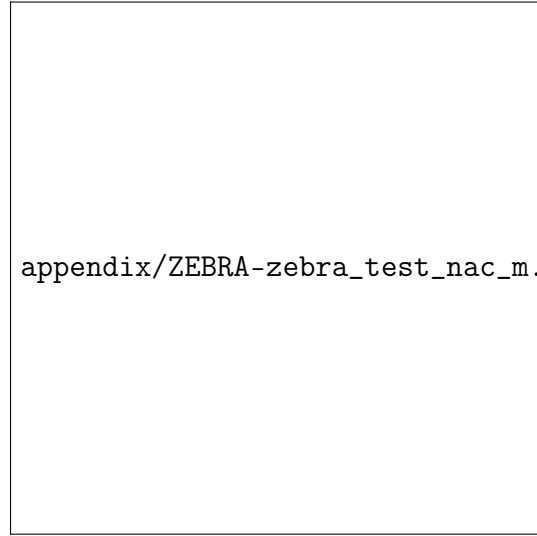


(b) NAC, dev-m

Figure 5: ZEBRA ECE curves for NAC on the development partitions.



(a) NAC, test-f

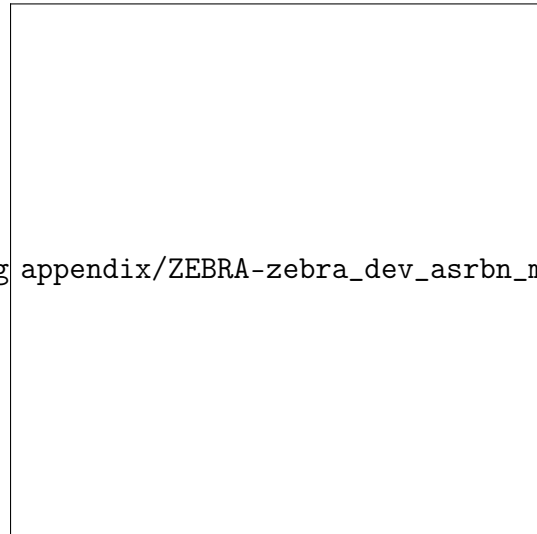


(b) NAC, test-m

Figure 6: ZEBRA ECE curves for NAC on the test partitions.



(a) ASRBN, dev-f



(b) ASRBN, dev-m

Figure 7: ZEBRA ECE curves for ASRBN on the development partitions.

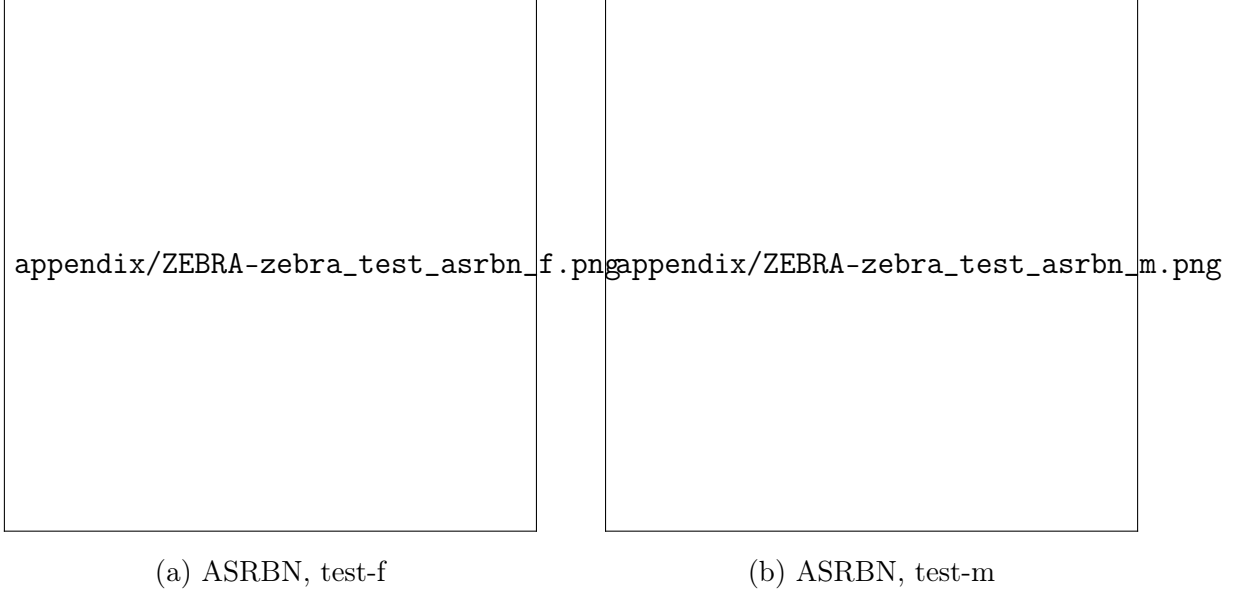
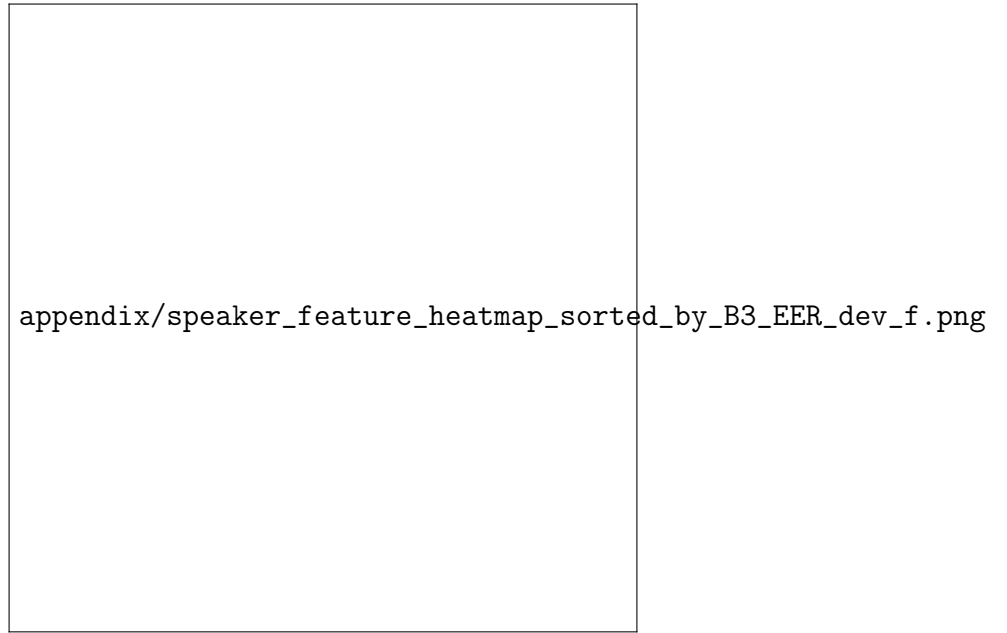


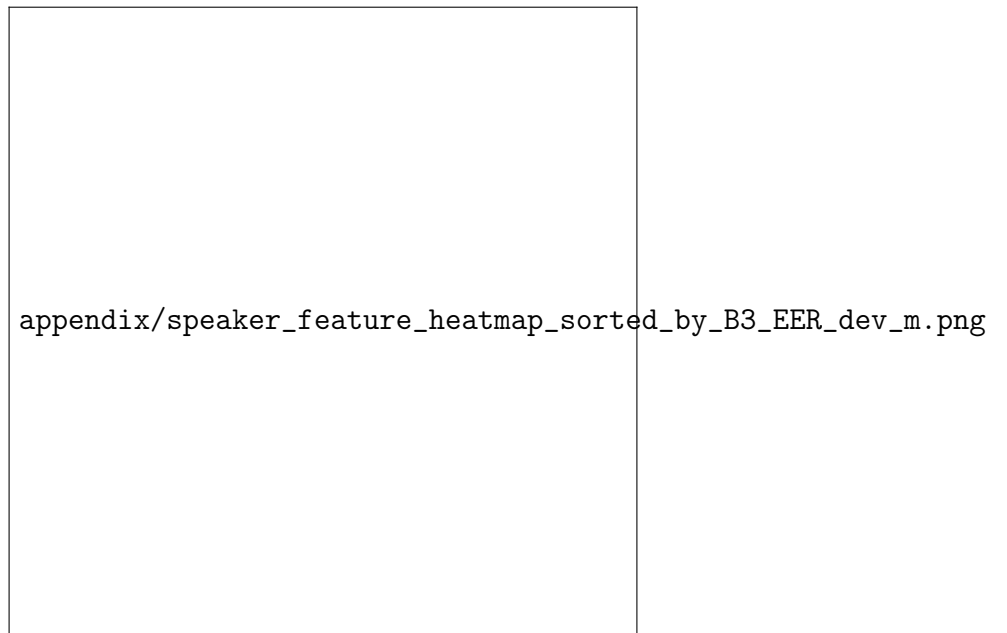
Figure 8: ZEBRA ECE curves for ASRBN on the test partitions.

A.2 Speaker-Level Feature Preservation Heatmaps

This section presents additional speaker-level heatmaps of feature preservation after anonymization. Speakers are sorted by EER, with lower values indicating higher vulnerability. Columns show standardized preservation scores across feature spaces, where warmer colors indicate above-average preservation and cooler colors indicate below-average preservation.

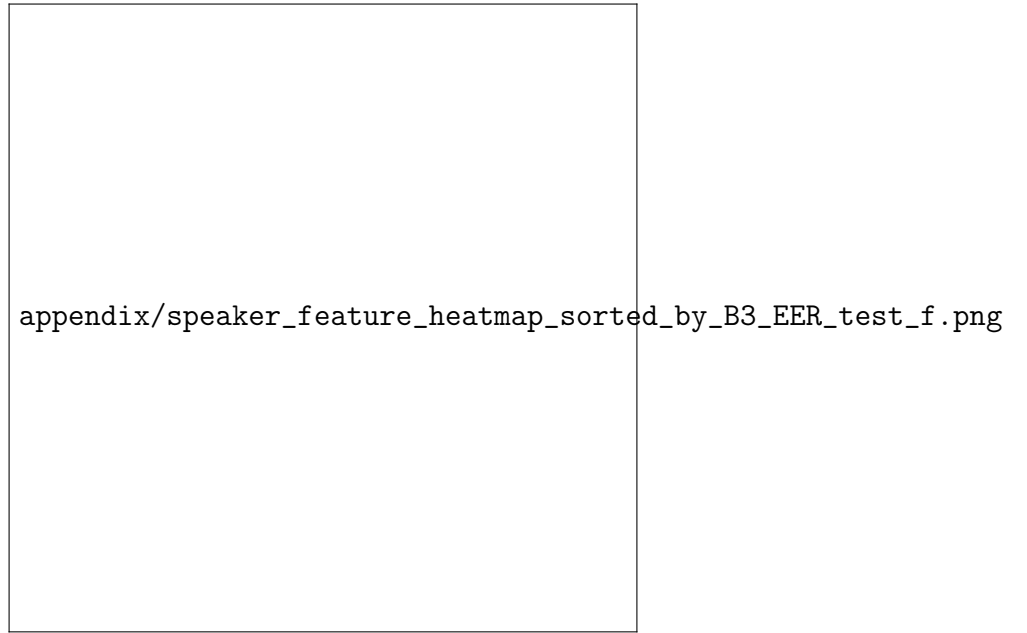


(a) B3, dev-f

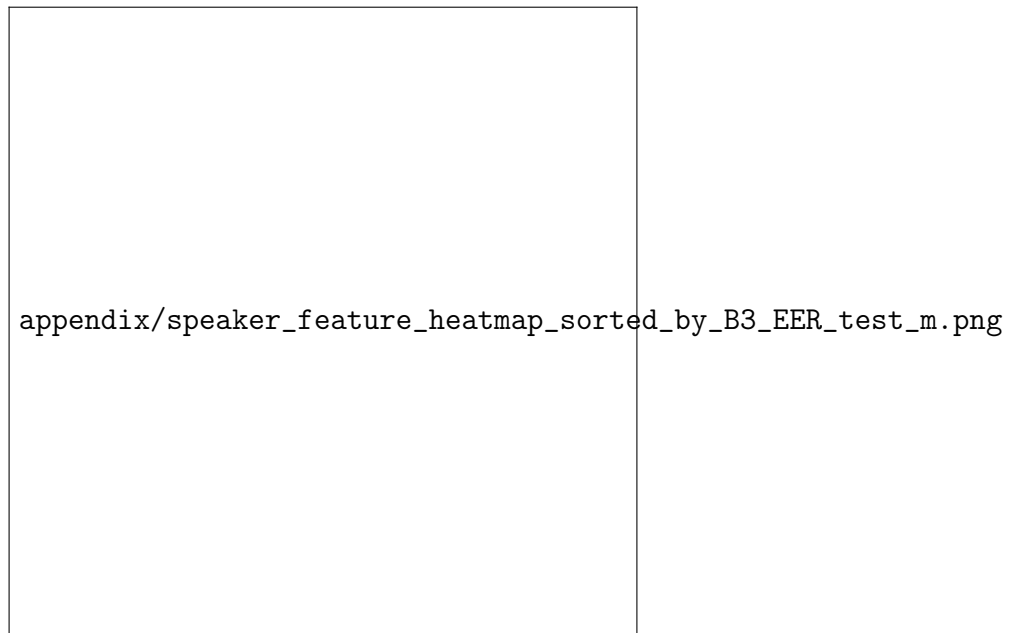


(b) B3, dev-m

Figure 9: Speaker-level preservation features sorted by B3 EER on the development partitions.



(a) B3, test-f

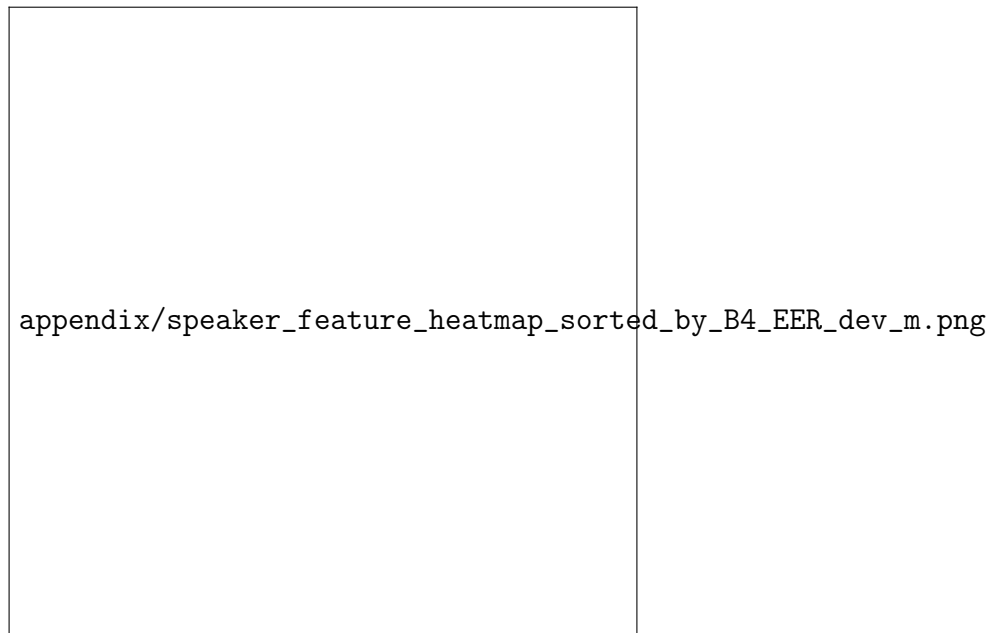


(b) B3, test-m

Figure 10: Speaker-level preservation features sorted by B3 EER on the test partitions.

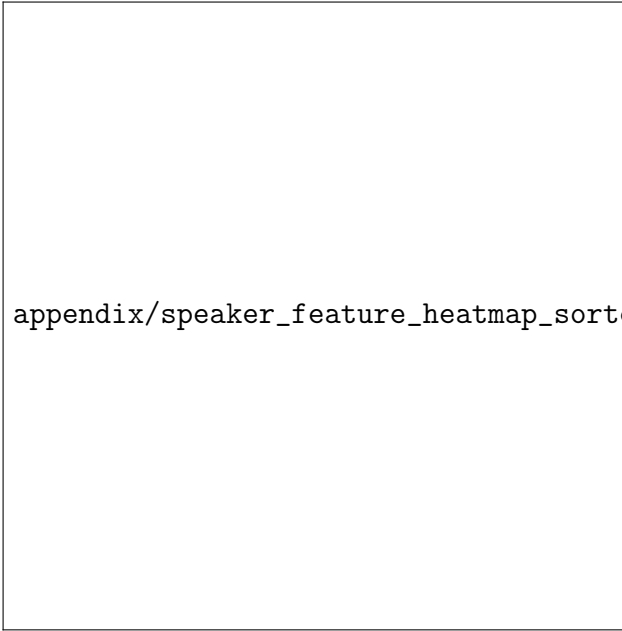


(a) B4, dev-f

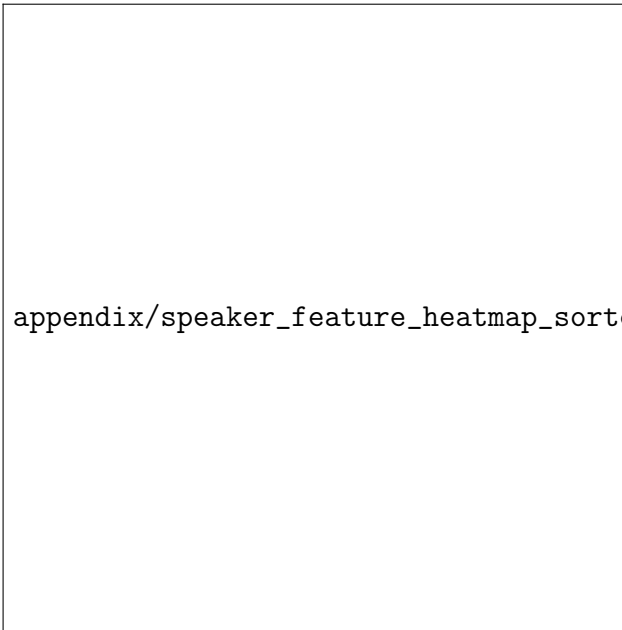


(b) B4, dev-m

Figure 11: Speaker-level preservation features sorted by B4 EER on the development partitions.

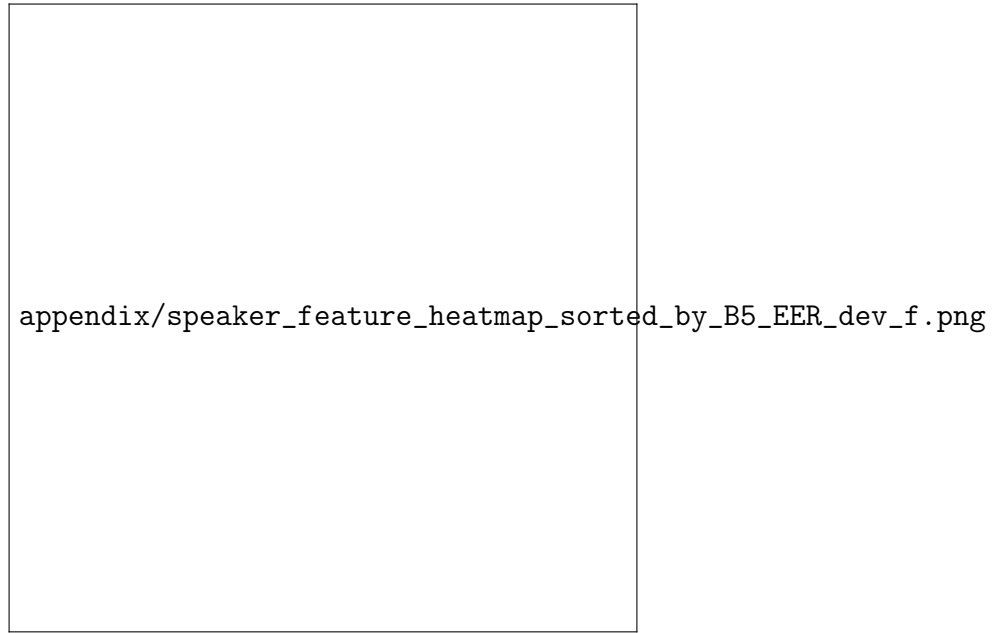


(a) B4, test-f

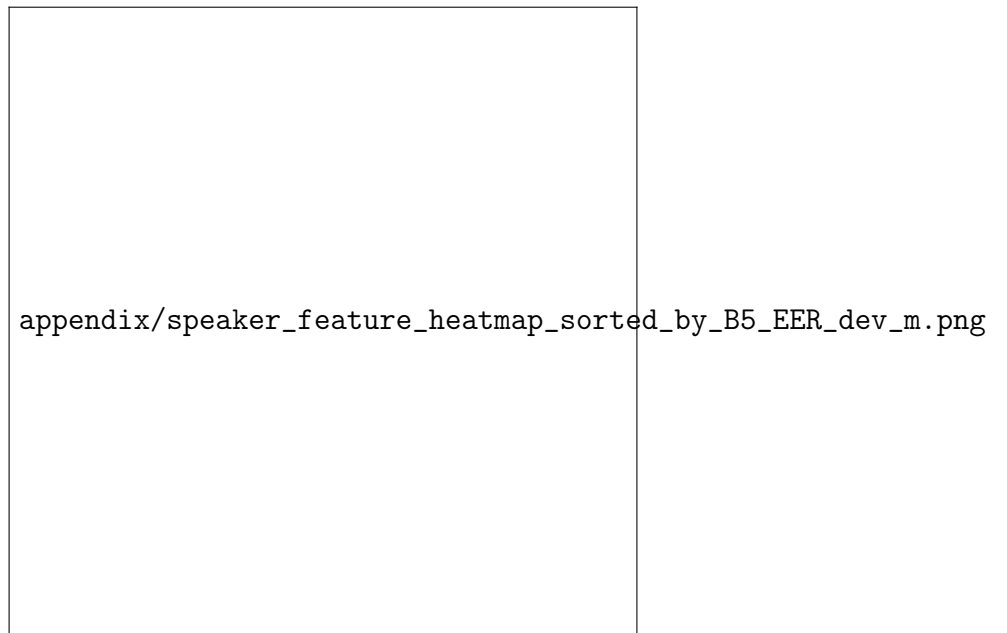


(b) B4, test-m

Figure 12: Speaker-level preservation features sorted by B4 EER on the test partitions.

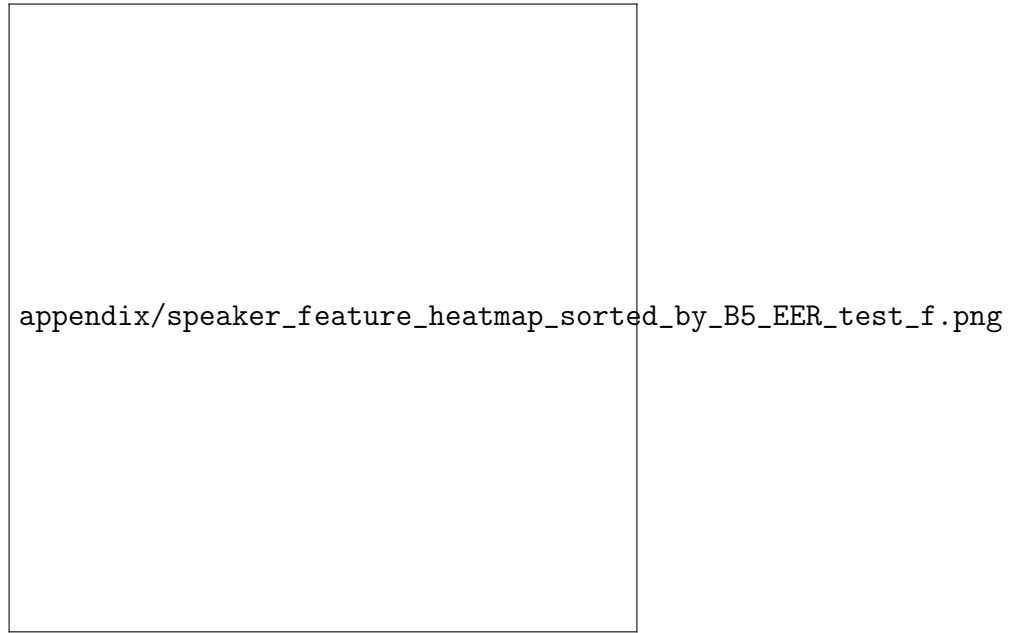


(a) B5, dev-f

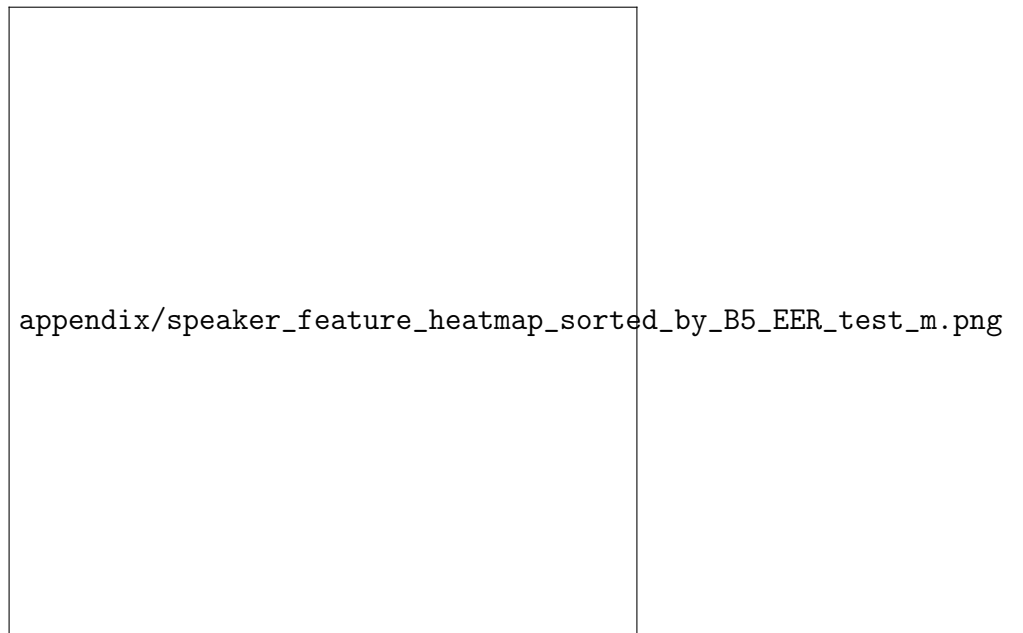


(b) B5, dev-m

Figure 13: Speaker-level preservation features sorted by B5 EER on the development partitions.



(a) B5, test-f



(b) B5, test-m

Figure 14: Speaker-level preservation features sorted by B5 EER on the test partitions.

A.3 Use of AI-Tools

The author of this thesis has used AI tools, i.e. ChatGPT versions GPT5.4 and GPT5.5, in order to assist in the code support when errors were encountered in the implementation of the evaluation metrics and in order to secure their compliance with their accompanied papers. AI was also used for the analysis part of the thesis, again with support in the implementation of the analysis code and the encountered errors. The author has also used this tool in order to compile the tables seen in this thesis and in some cases in order to refine the text. All the other parts of the thesis, including the writing, the background work, the related work, the analysis and the interpretation of the results, were done solely by the author.